

AKADEMIKA

RFL-AMS-A
ANTENNA MEASUREMENT SYSTEM
EXPERIMENTAL MANUAL

....A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. ADVANCED DIGITAL COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

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SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting Cords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting Cords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions' sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions' judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

Sr. No	Chapter	Page No.
1	Introduction	6
2	Functional Block	8
	A. Front panel Description	9
	B. Horizontal Assembly Description	10
	C. Vertical Assembly Description	11
	D. Vertical Mount Assembly (optional)	12
3	Technical Specifications	14
	A. Lists of Antenna	16
	B. Deliverables	17
4	Software Installation	18
	A. Operation Modes	21
	B. Software Control Mode	24
	C. Panel Control Mode	28
5	EXPERIMENT NO. 01 Measure the variation of field strength /inverse square law	35
6	EXPERIMENT NO. 02 Prove the reciprocity theorem of antenna.	39
7	EXPERIMENT NO. 03 Plot Radiation pattern of all Wired Antenna	43
8	EXPERIMENT NO. 04 Plot Radiation pattern of all Aperture Antenna.	53

9	EXPERIMENT NO. 05	57
	Plot Radiation pattern of all Reflector Antenna.	
10	EXPERIMENT NO. 06	61
	Plot Radiation pattern of all Array Antenna.	
11	EXPERIMENT NO. 07	65
	Measurement in co-polarization & cross polarization	
12	EXPERIMENT NO. 08	69
	Circularly Polarized Antennas	
13	EXPERIMENT NO. 09	73
	Planar (Micro strip) antenna category:	
	A. Plot Radiation pattern of all Planar (Microstrip) Antenna.	74
	B. Design and Study of Basic Micro strip Shapes and its configuration.	80
	C. Effect of different feeding techniques in Micro strip Antenna.	85
14	EXPERIMENT NO. 10	89
	Gain Measurement of RMSA Antenna	
15	EXPERIMENT NO. 11	93
	Calibration of Planer Dipole Antenna	
16	EXPERIMENT NO. 12	97
	User defined antenna pattern measurement	
17	Annexure 1	102
	Antenna Calculation Formulae	
18	Annexure 2	103
	Antennas & Their Center Frequencies	

INTRODUCTION

Antennas are a fundamental component of modern communications systems. By definition, an antenna acts as a transducer between a guided wave in a transmission line and an electromagnetic wave in free space.

Why Antenna Measurements?

A graphical representation of the relative distribution of the radiated power in space is called a radiation pattern. The radiation pattern of the antenna is of principle concern when engineering a communications system. Let's assume that a signal needs to be sent from an antenna on the ground to a satellite in orbit. This would require a radiation pattern with the majority of its radiated power focused into orbit. If the antenna is not engineered to do so, contact cannot be established between the signal source and its target. There are many different ways to manipulate a radiation pattern to meet the demands of a specific task. These concepts are the principle focus of this lab assignment. Implementing this lab assignment, students will examine the radiation patterns of several antennas by hands on field testing.

Antennas demonstrate a property known as reciprocity that is an antenna will maintain the same characteristics regardless if it is transmitting or receiving. When a signal is fed into an antenna, the antenna will emit radiation distributed in space a certain way.

Motivation

Only the most fundamental antennas were chosen for this lab assignment. This allows us to see visually how the most common types of real-world antenna designs function.

Many antenna systems courses introduce theory and systems characteristics without a laboratory component. Since the development of antenna systems benefits from a physical understanding of wave propagation and competent measurement techniques, laboratory experiments should be an integral part of the lecture course.

To address this issue, a complete laboratory course involving antenna measurements using Spectrum analyzer antenna measurement system has been developed



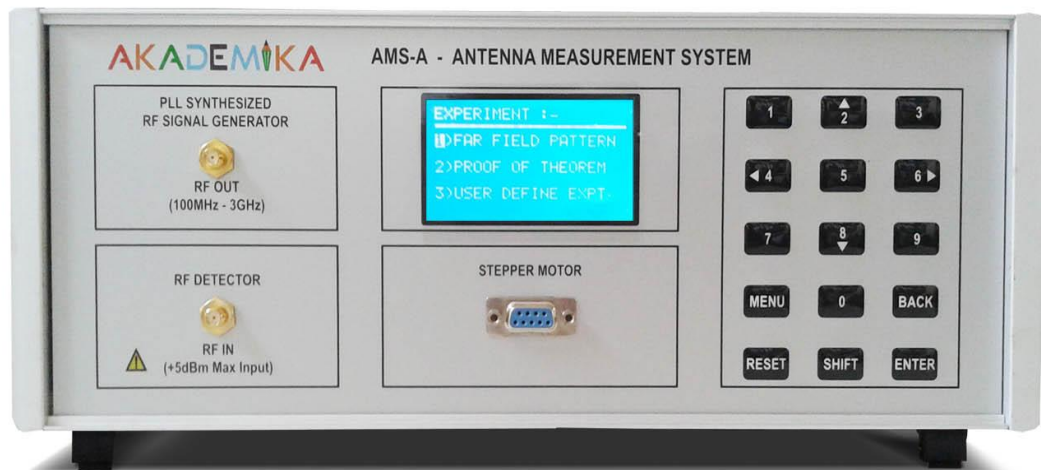
Features of Antenna Trainer System

- Covers UHF, L, S Band Wide frequency coverage up to 3 GHz
- PLL Synthesized Source and Detector working up to 3GHz with high dynamic range of power Transmission
- Customized selection of antenna from the list as per syllabus requirement
- Practical approach for Microstrip Antenna design covering concepts of size reduction, Bandwidth enhancement, stacked Multilayer configuration, impedance matching
- Non conductive and non magnetic Transmitter and Receiver pattern plotting software
- Single training system to teach all types of antenna measurement

FUNCTIONAL BLOCK



AMS-A FRONT PANEL DESCRIPTION



1. RF OUT

This is a PLL synthesized RF signal generator, the signal which is going to transmit will be available from this generator. The output frequency is from 100 MHz to 3GHz. Its output impedance is 50 ohms. The transmitting antenna can be connected to it using the SMA-SMA MALE cable provided. The power can be varying between -4dBm to +5dBm. An external attenuator is not needed.

2. RF IN

This is RF signal DETECTOR, where the RECEIVED signal from antenna should be connected. The RF IN terminal should connect to receiver antenna through cable. Its impedance is 50 ohms. The receiving antenna can be connected to it using the SMA-SMA MALE cable provided. The power can be vary between -4dBm to +5dBm. An external attenuator is not needed. Inbuilt software controlled attenuator is present.

3. LCD

This is a TFT LCD Display, used to display the MENU options as well as most of the selections done by the user i.e. frequency, rotation angle, received power in dBm, instructions and message is displayed.

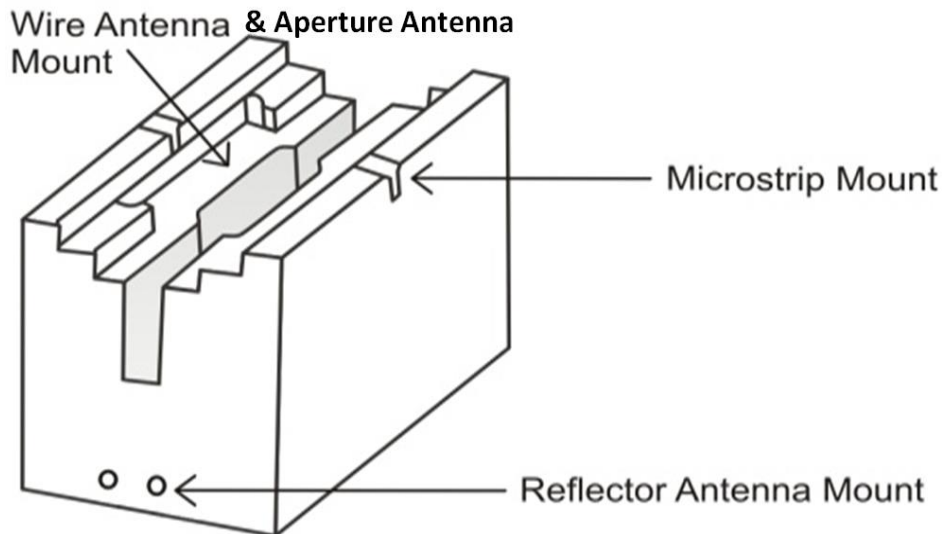
4. KEYBOARD

The keyboard is provided to give input to trainer.

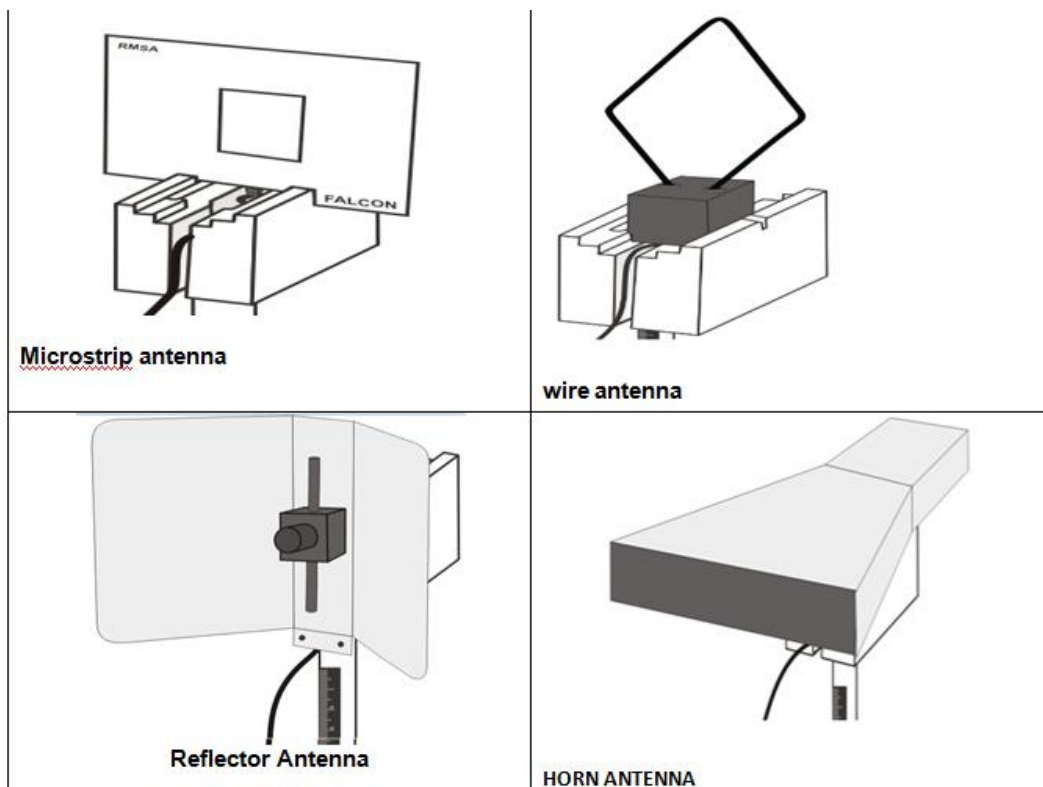
- MENU: by pressing this key the menu options are available.
- BACK: by pressing this key a user can move one step backward.
- ENTER: when a user wants to confirm its selection, ENTER key should be pressed.
- SHIFT: this KEY is used to enable the UP-DOWN arrow keys.
- RESET: by pressing this key the trainer gets reset.

HORIZONTAL ASSEMBLY DESCRIPTION

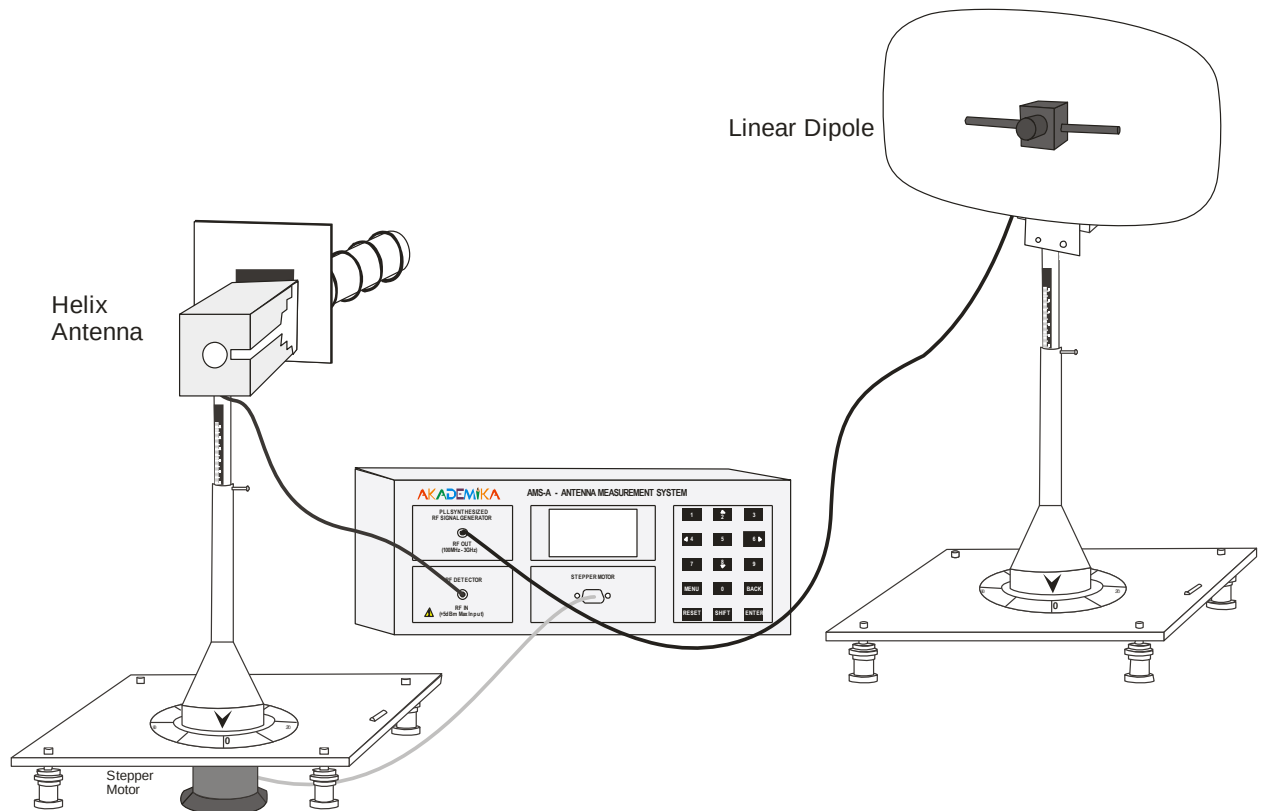
The below figure shows horizontal mount which is made up of polyxeamethalene polymer. This is universal mount to support transmitting and receiving antenna. Appropriate slots are provided to fit the antenna assembly. Press fit facility is available for Wire, aperture and planner antenna. The reflector antenna is required screw fitting.



The below examples are showing arrangement of antenna assembly with its appropriate fittings.

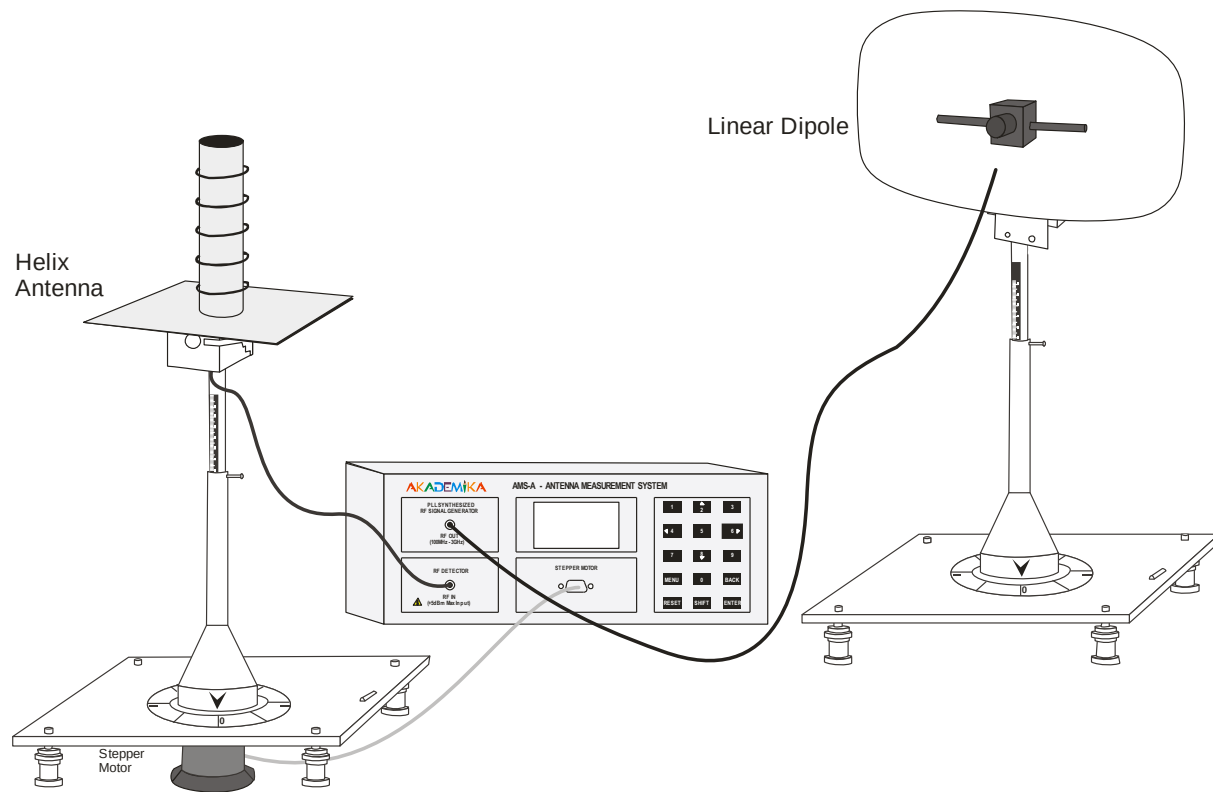


1. Attach four height adjuster to base plate
2. Check the spirit level; adjust the height of the supporter until the spirit bubbles comes in the center.
3. Press fit the conical mount to base plate. To reduce the friction the bearing is present.
4. Press fit Hollow rod to conical mount
5. Fit the horizontal mount to solid rod, rotate the mount clockwise so threads are fit with solid rod.
6. Insert the above assembly in hollow pipe



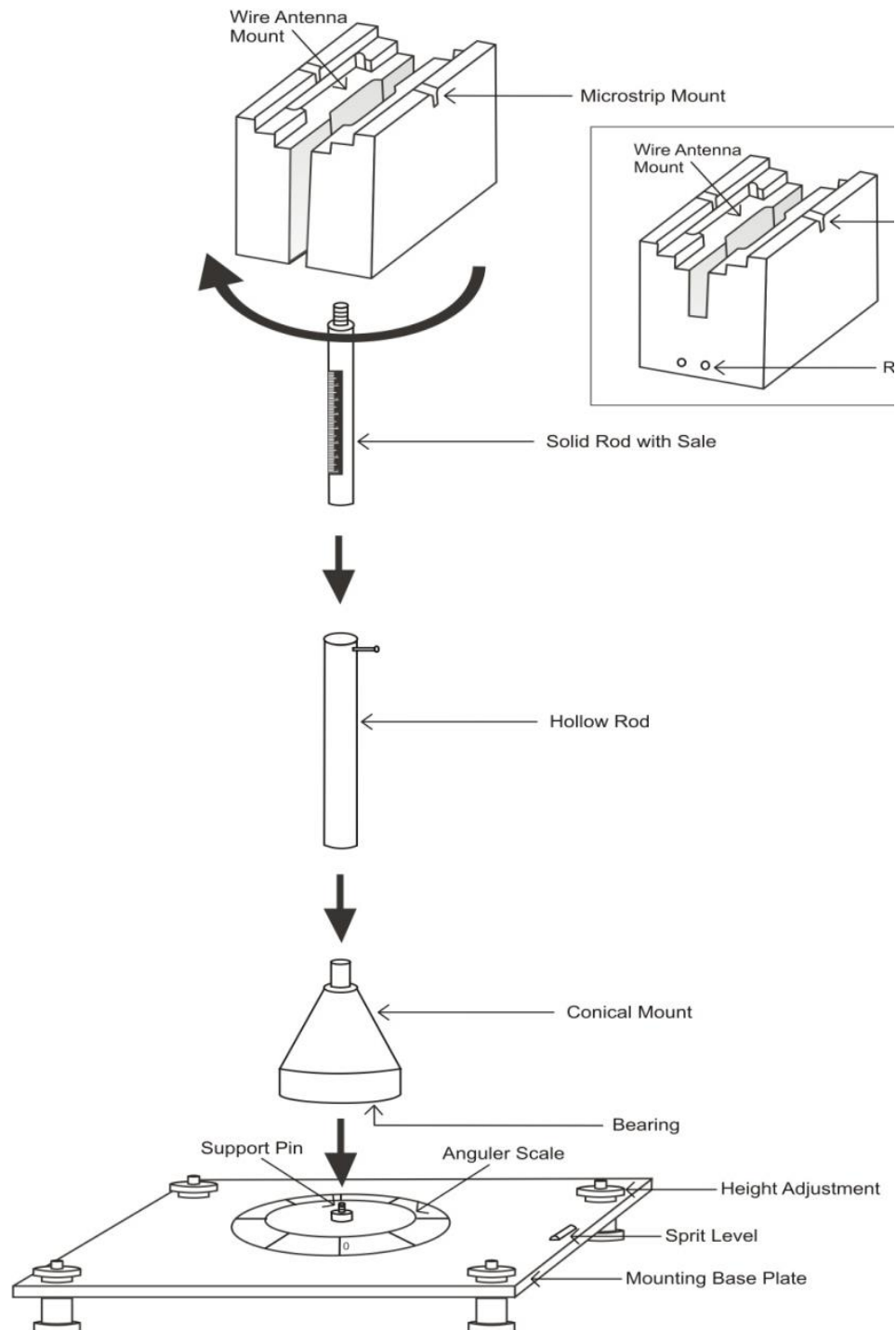
VERTICAL ASSEMBLY DESCRIPTION

1. Attach four height adjuster to base plate
2. Check the spirit level, adjust the height of the supporter until the spirit bubbles comes in the center.
3. Press fit the conical mount to base plate. To reduce the friction the bearing is present.
4. Press fit Hollow rod to conical mount
5. Fit the vertical mount to solid rod, rotate the mount clockwise so threads are fit with solid rod.
6. Insert the above assembly in hollow pipe
7. Adjust the solid rod in such a way that scale is readable while measurement, Initially do not provide height to solid rod



Vertical Mount Assembly (optional)

Antenna measurement system consists of vertical mount facility to provide radiation pattern analysis. Vertical mount assembly consist of base plate, vertical rod, stepper motor, universal mount. Antenna assembly can fit in the universal mount.

ASSEMBLY INSTRUCTION

TECHNICAL SPECIFICATIONS

RF Source	
Source types	PLL Synthesized based integrated with VCO
Frequency Range	100MHz to 4GHz
Frequency Resolution	100KHz
Dynamic range of transmitter	85 dBm
Transmitted power min	-4dBm (103dBuV)
Transmitted power max	+5dBm (112dBuV)
Output-frequency switching time	10ns
Tuning range	4.4GHz
Phase noise	-90dBc/Hz
Normalized Phase Noise Floor	-220Dbc/Hz
Spurious Signals Due to PFD Frequency	80 dBc
RF Detector	
Detector type	Logarithmic Detector
Frequency Range	1 MHz to 8 GHz
Resolution	0.1 dB
Dynamic range	85dB
Response time	6 ns/8 ns (fall time/rise time)
Stability over temperature	±0.5 dB (−40°C to +85°C)
Noise level	<-120dbm
Impedance	50ohm
LCD Display	20 character X 4 Line with backlit
Interface Cables	
Connection	Cable Type
1.Trainer board to TX antenna	RG317 SMA to SMA Male
2. Trainer board to RX antenna	RG317 SMA to SMA male
3. Antenna trainer module to PC	USB to USB cable
4.Trainer supply	Standard power supply cable 230v
Software	
	Radiation pattern plotting and analysis software suitable for windows environment

Antenna Mounting Assembly	
Antenna Positioner	• The Transmitter and Receiver Antenna stand is made of special material which is inert to EM frequency and it has engraved height and angle scale on it. It has facility to adjust the height and level. • Universal plug and fix Antenna mounts are provided to hold the antenna assembly in vertical and horizontal orientation for co and cross polarization. • Height scale for height adjustment, base plate height adjustment with spirit level.

LIST OF ANTENNA

Sr. No	Type of Antenna	Name of Antenna	Qty
1	Wire Type	Wire Dipole	2
2		Helix	1
3		Yagi-Uda 5 element	1
4		Rectangular Loop	1
5		Vee Dipole	1
6		Folded dipole	1
7		Cross Dipole	1
8		Monopole	1
9	Aperture Type	Open ended waveguide	1
10		E-Horn	1
11	Planner Type	RMSA	1
12		TMSA	1
13		CMSA	1
14		Insert Feed Patch	1
15		RMSA Circular Polrized	1
16		Annular ring	1
17		2x1 Patch Array	1
18		Chip	1
19		Monopole	1
20		Planner Dipole	2
21	Array	Collinear array	1
22		End fire	1
23		Broadside array	1
24	Reflector Type	Corner Reflector with feeder & adapter	1
25		Parabolic Reflector with feeder, adapter	1

AMS-A Deliverables

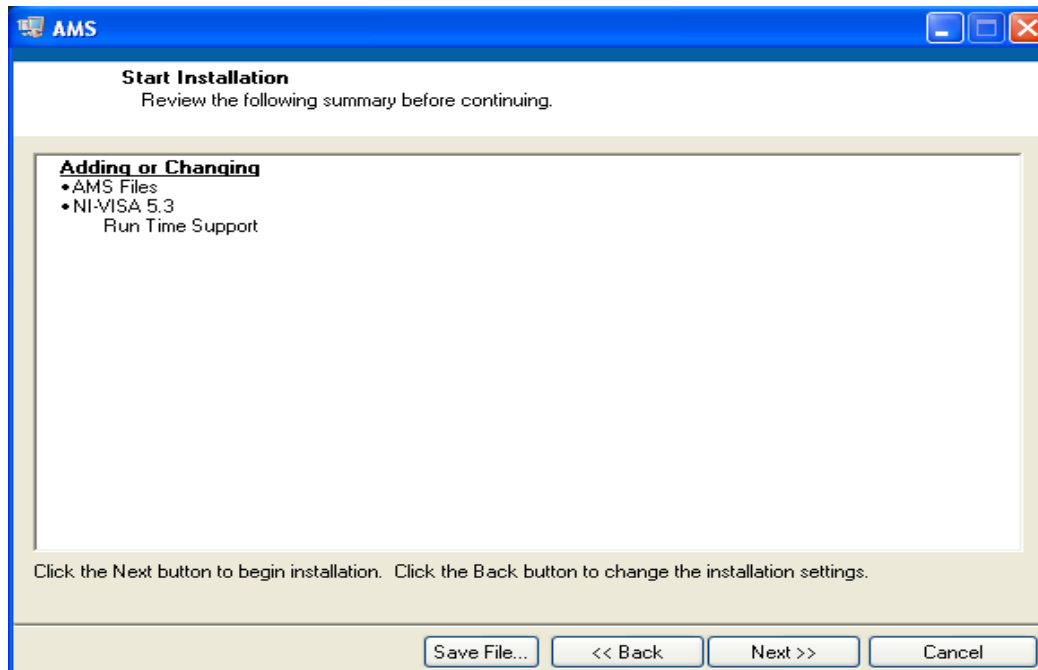
Sr. No.	Description	Quantity
1	Antenna Measurement System	1
2	SMA TO SMA MALE ADAPTER	1
3	ANTENNA MOUNTING ASSAMBLY	
	A. Height Adjuster Legs	8
	B. Base plate for Transmitter	1
	C. Base plate for Receiver with Stepper Motor & Conical Mount	1
	D. Conical mount for Transmitter	1
	E. Hollow rod	2
	F. Solid rod with scale	2
	G. Big Nut for Solid rod of Transmitter	1
	H. Universal Mount with Screws	2
	I. Screw Driver	1
	J. Allen Key	1

Accessories

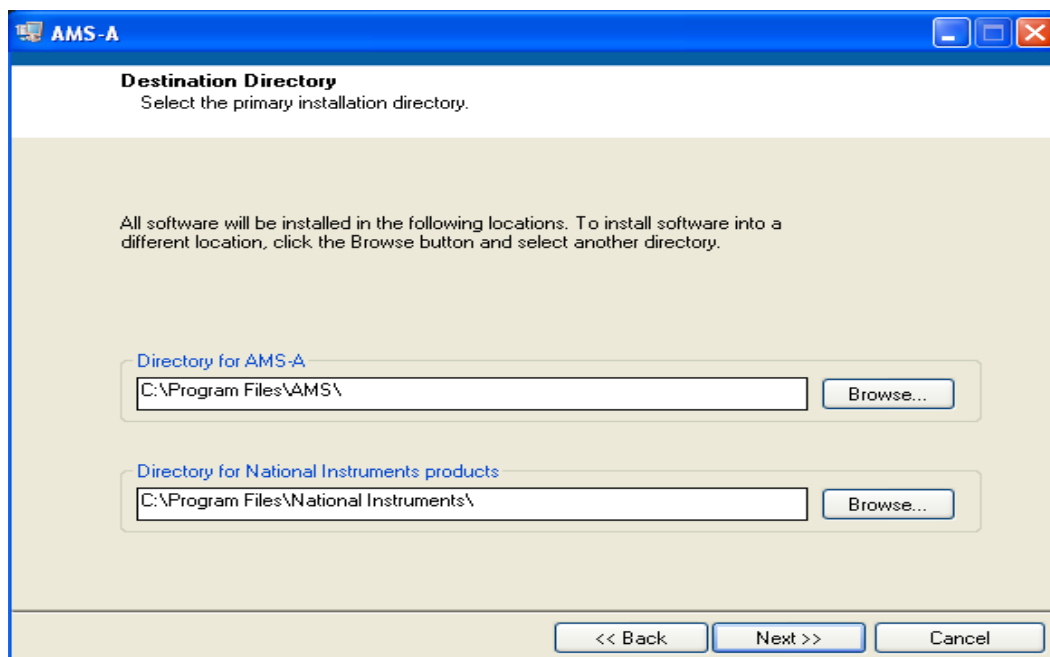
1	Power Cord	1
2	USB A male To USB A male cable	1
3	SMA male to SMA male 2m cable	1
4	SMA male to SMA male 1m cable	1
5	9 pin D type male to female cable	1
6	RFL-AMS-A Expt Manual	1
7	Pendrive	1

AMS-A Software Installation

- 1) a) Run the setup.exe available from the AMS installer folder (*CD provided with kit*), then following window will appear.



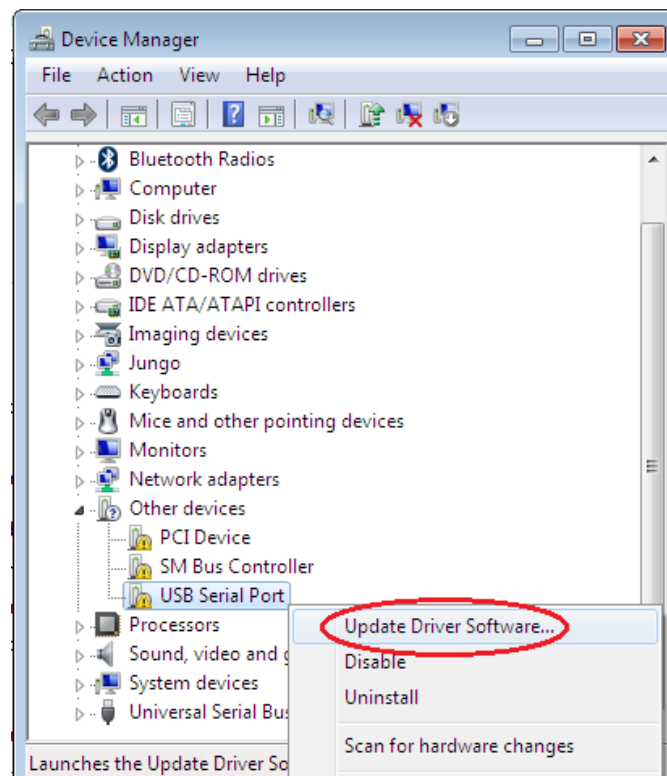
Click on Next then.



- b) Then select the directory where you want to install the setup, then click on Next.
c) Then click on finish.

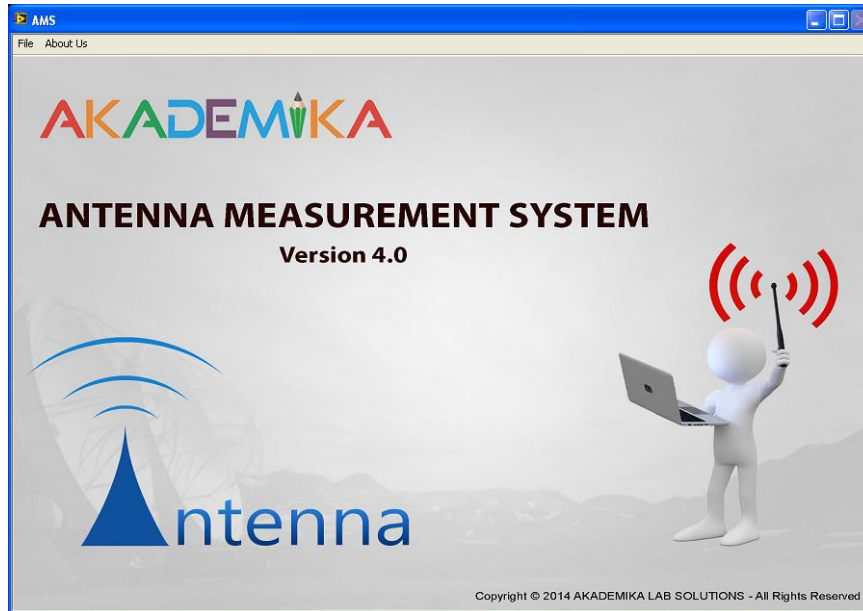
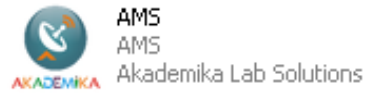
- 2) Restart the PC.

- 3) Connect the **AMS (TRAINER KIT)** to the **PC** using USB to USB cable (*provided with the kit*). Sometime FTDI Drivers get installed automatically or sometime we you have to install them.
- 4) For installation of FTDI Drivers manually follow the procedure:
 - Right click on **My Computer**
 - **Properties**
 - **Device Manager**
 - **Others**
 - Right Click on **USB Serial Port** & select **Update Driver Software**



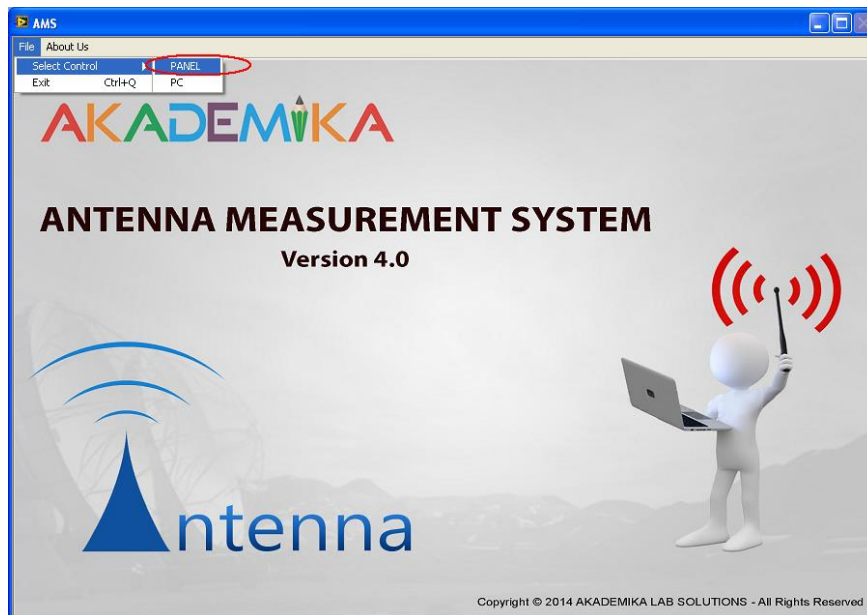
- **Browse my Computer for Driver Software**
- Select proper folder for Driver software from CD provided with kit
- **Click Next**
- Follow the same procedure once again then you will get **USB Serial Port COM** Address.

- 5) After the installation, go to directory where software installation is taken place and open the **AMS-A**

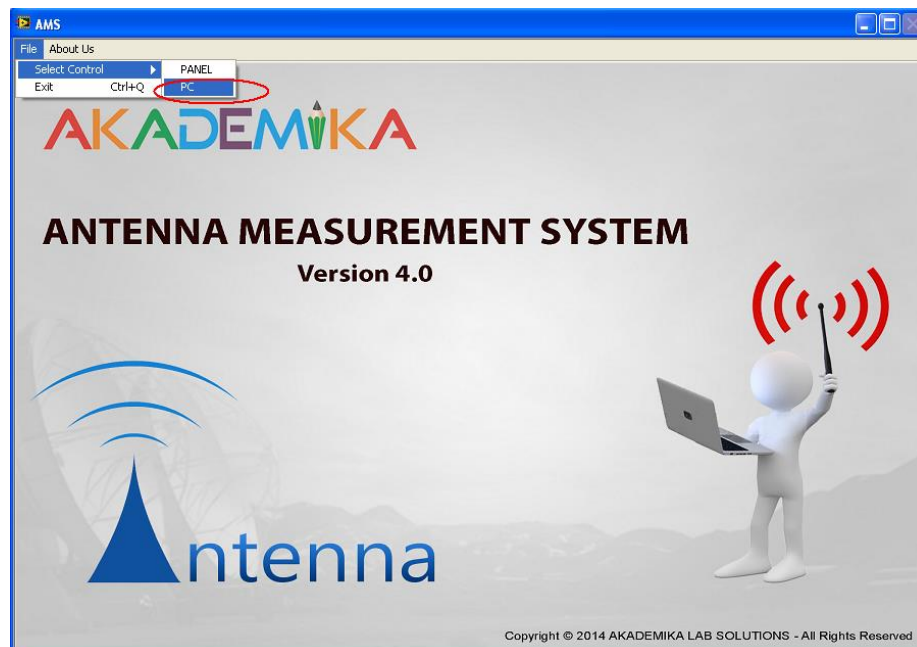


OPERATION MODES

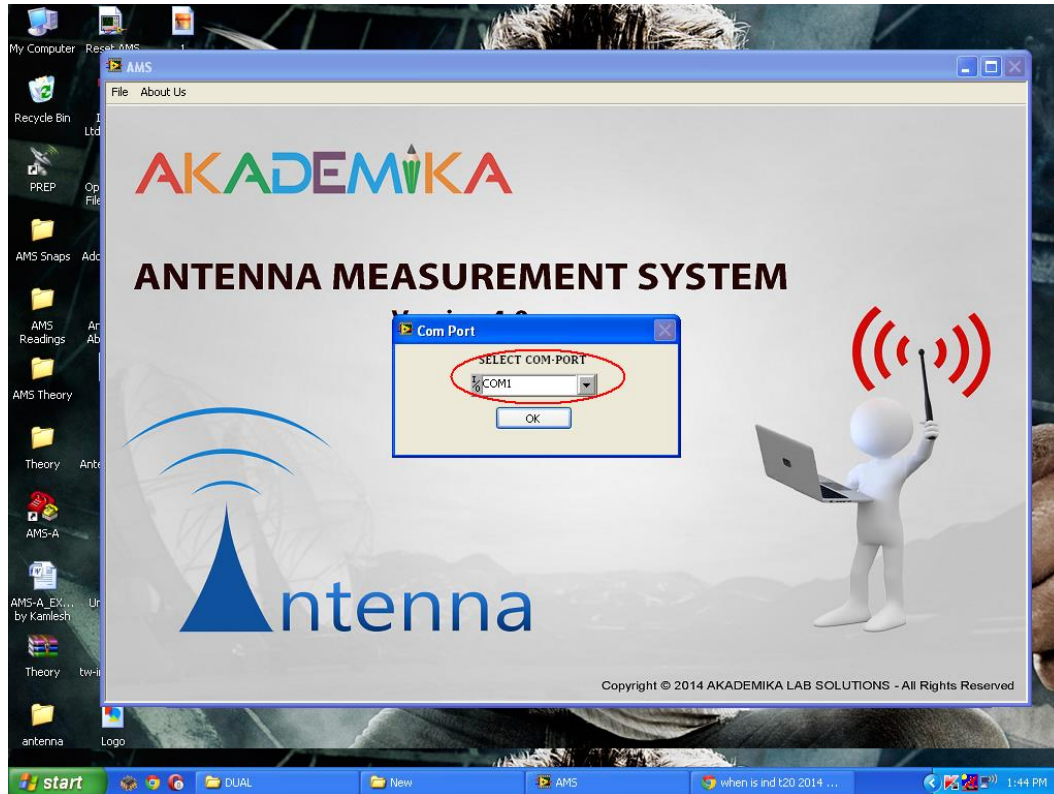
Select **File: Select Control: PANEL** for hardware control/panel mode .



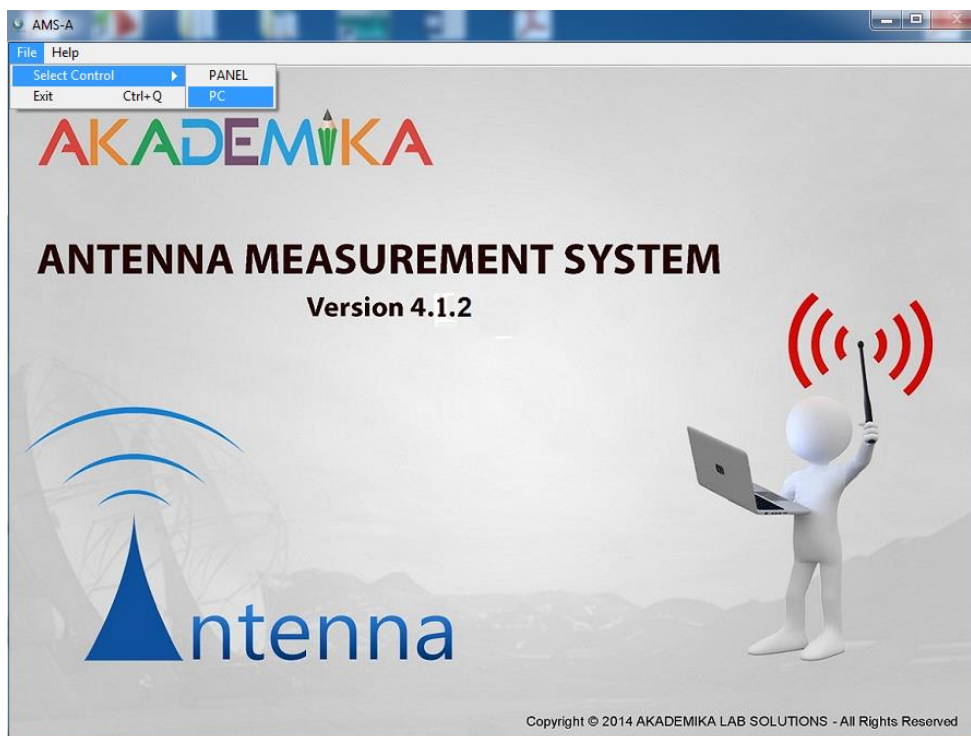
Select **File: Select Control: PC** for Software Control mode.



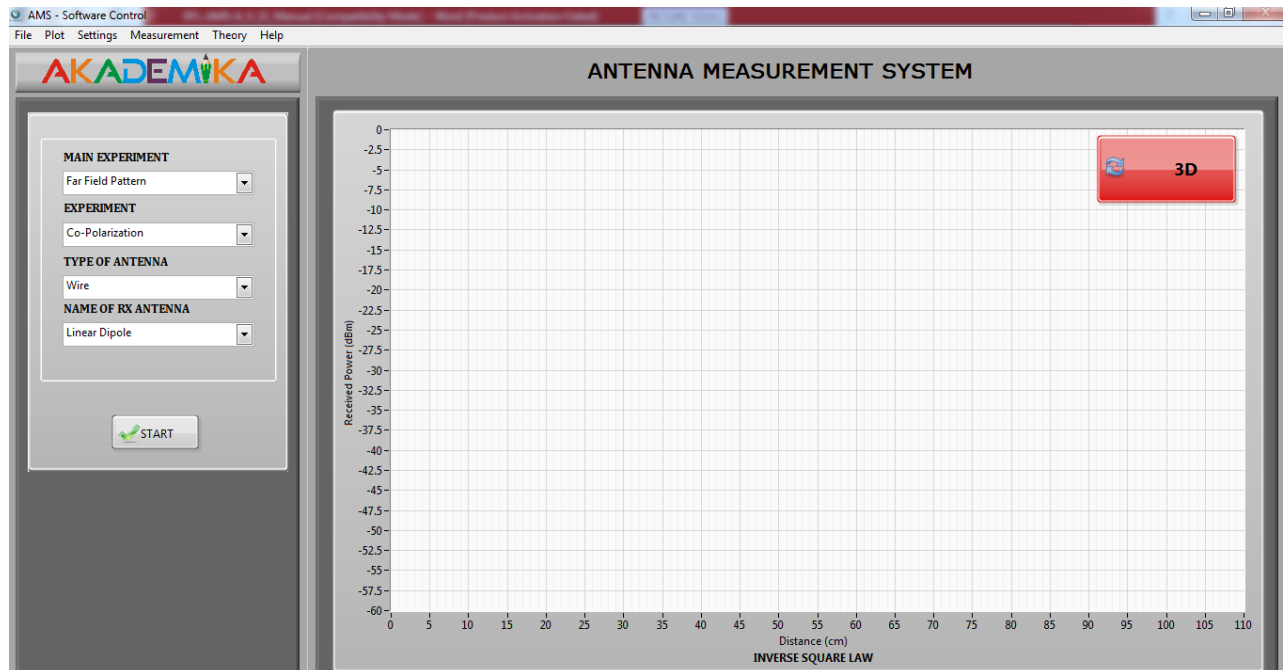
1. Select COM PORT.



2. Two respective windows will be opened for each control mode, as shown below.
Panel control mode / PC Control mode:



Software PC Control Mode :

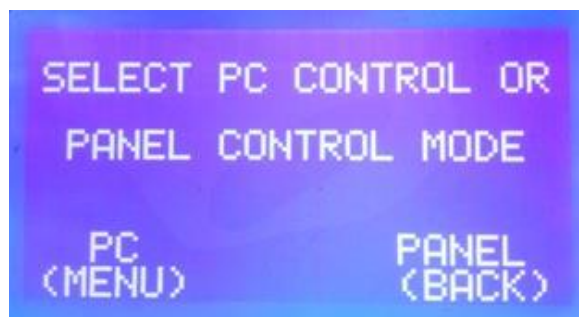


A. Software PC Control Mode

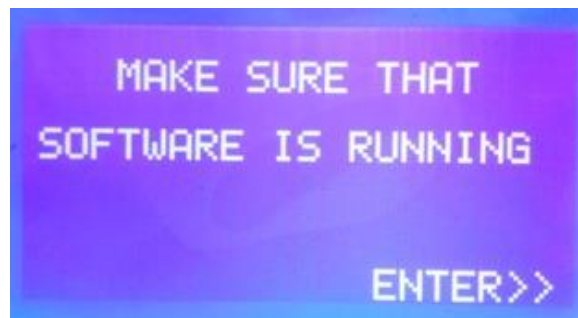
1. On the **AMS Trainer Kit**, Press **Menu** button on Front Panel:



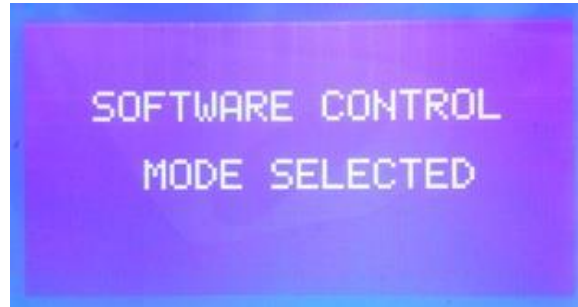
2. Select Control Mode:



3. Make sure that Software is running and press **ENTER**.



4. A continuous Window will be appeared throughout.



5. Press MENU key for PC control mode.

Run the Software in PC control mode as discussed earlier.

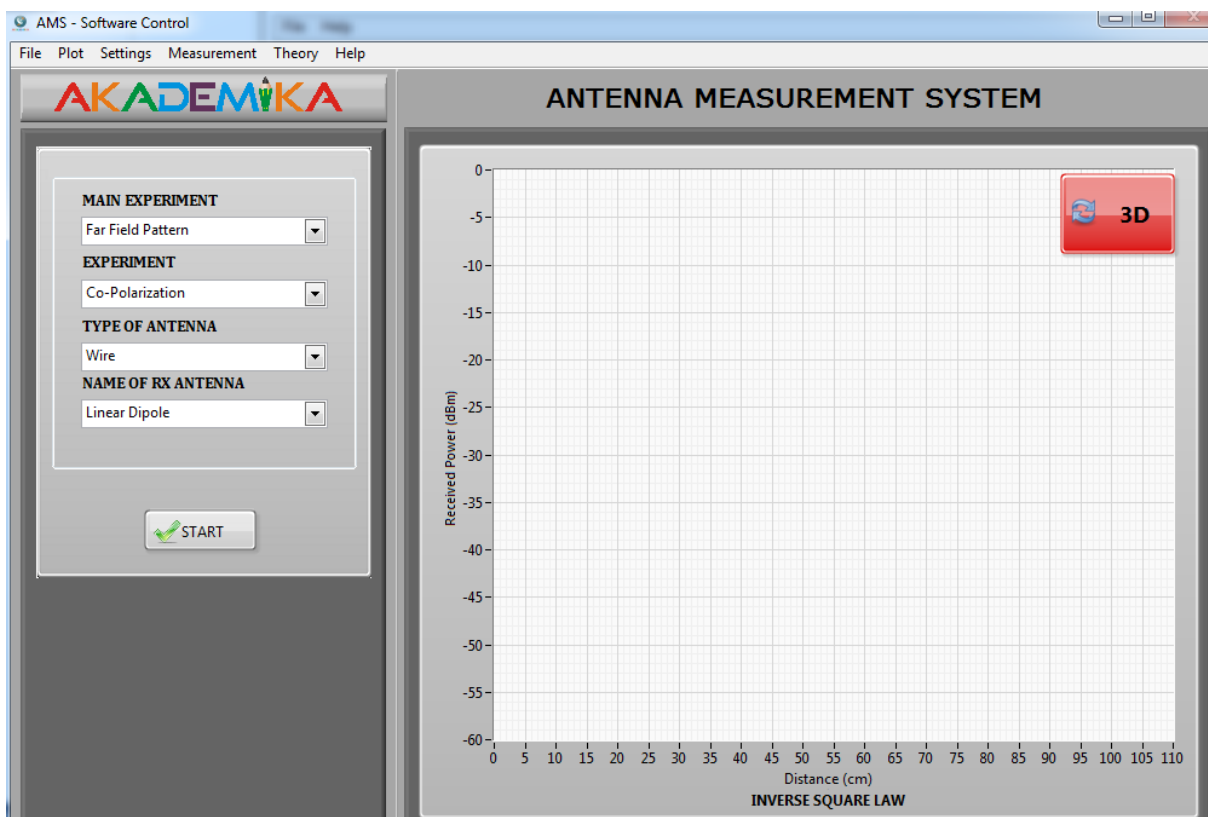
Select – **Main Experiment**

Experiment

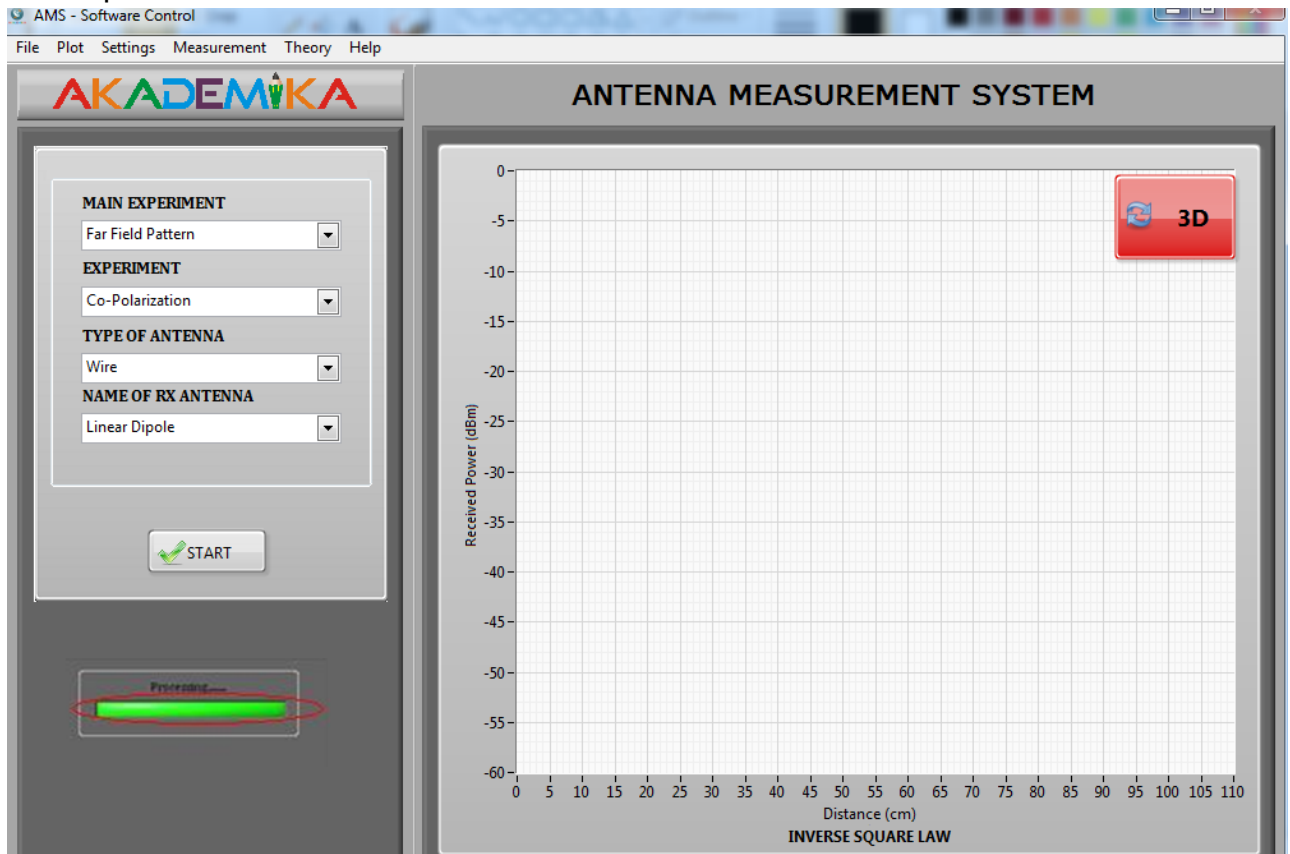
Type of Antenna

Name of Receiver Antenna

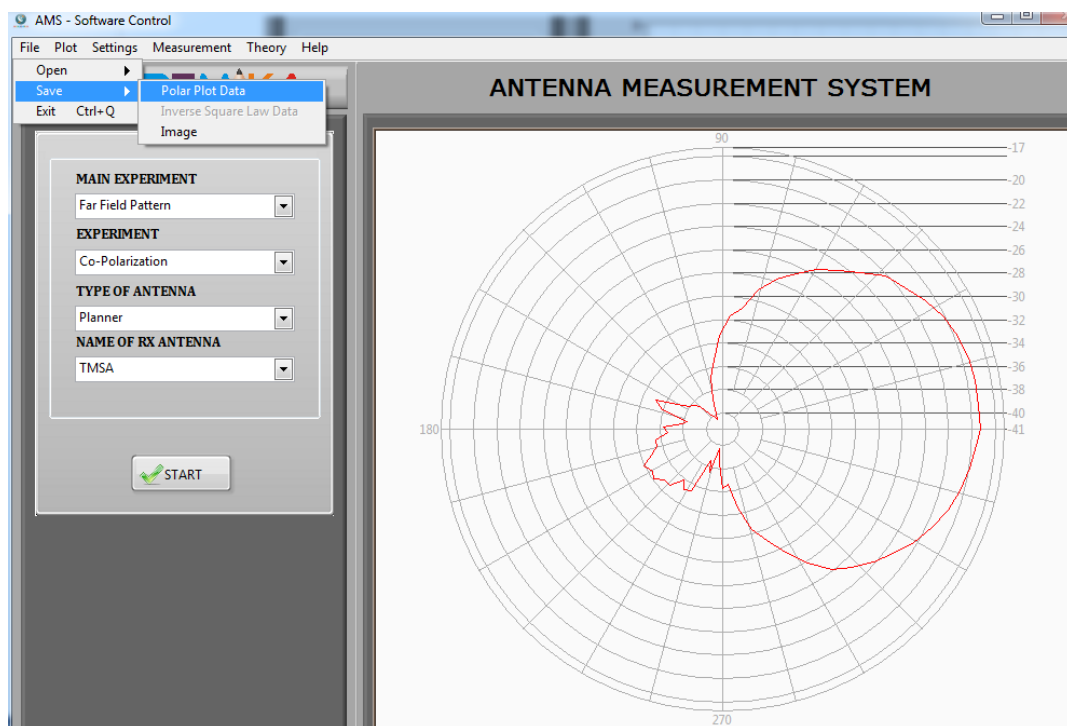
Click on **START**.



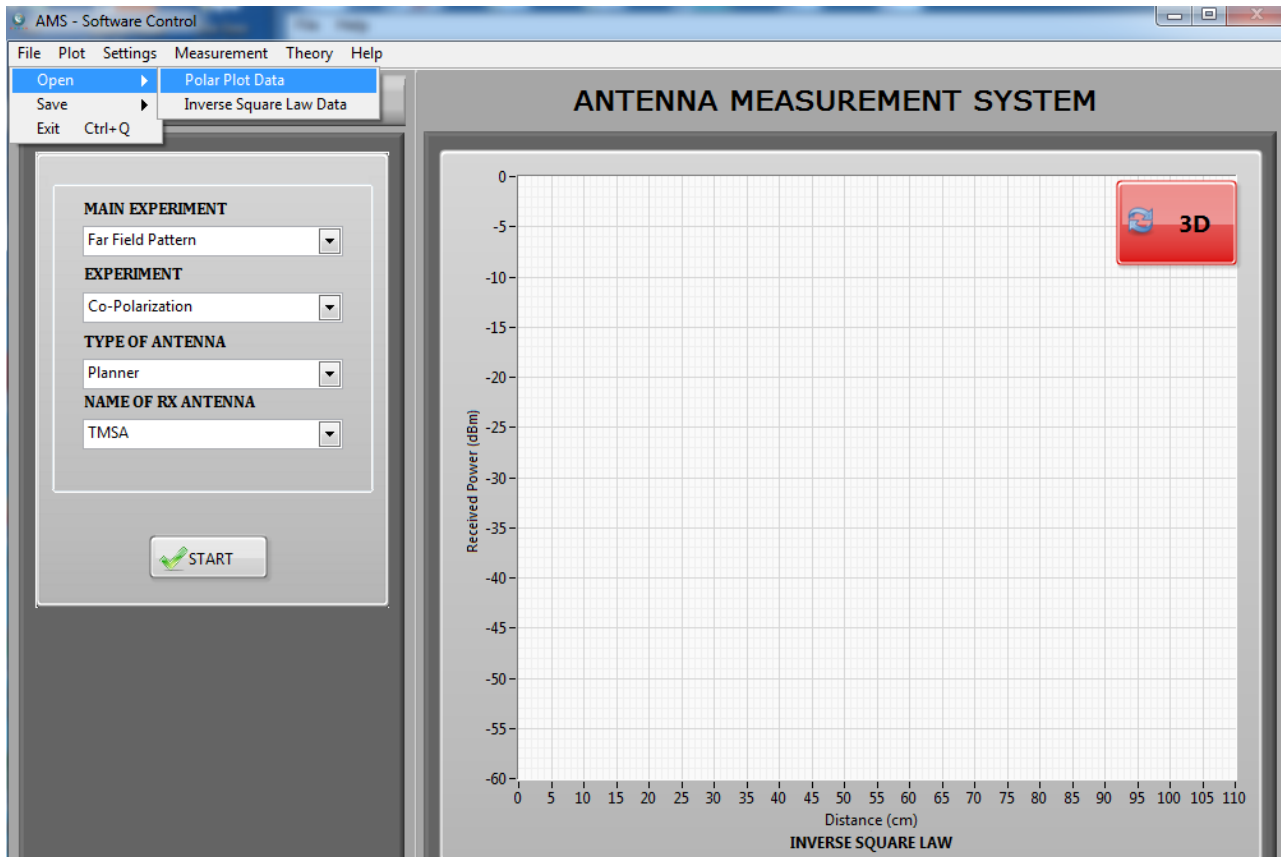
6. Start the Experiment. Processing bar will be appeared until the Plot has been plotted.



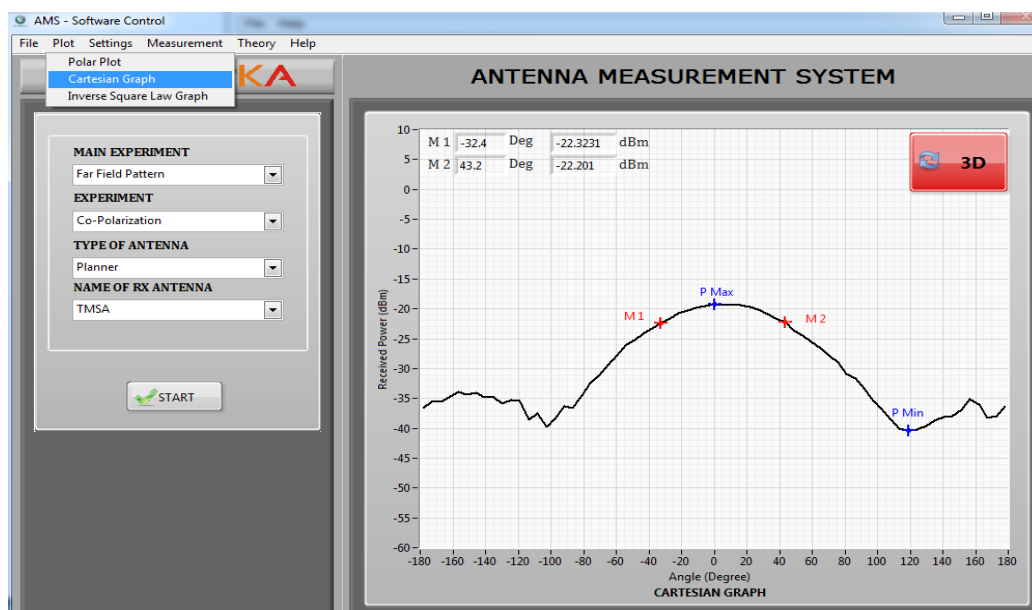
7. Plot will be plotted. You can save Plot image (.PNG format) and Readings (.CSV format) which can be viewed in MS EXCEL) from Menu-
File: Save:Polar Plot Image & File:Save:Readings.



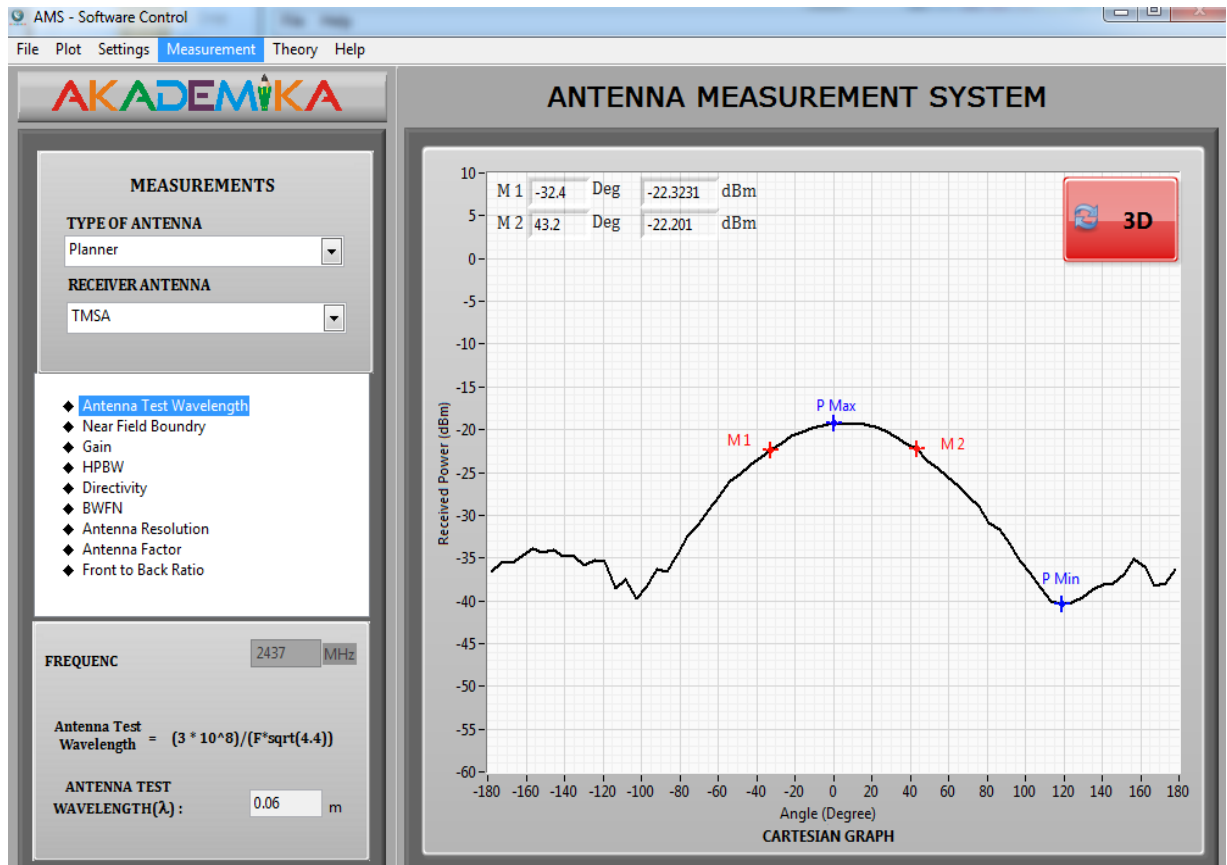
8. You can Open the Readings from Menu - **File: Open: Polar Plot Readings.**



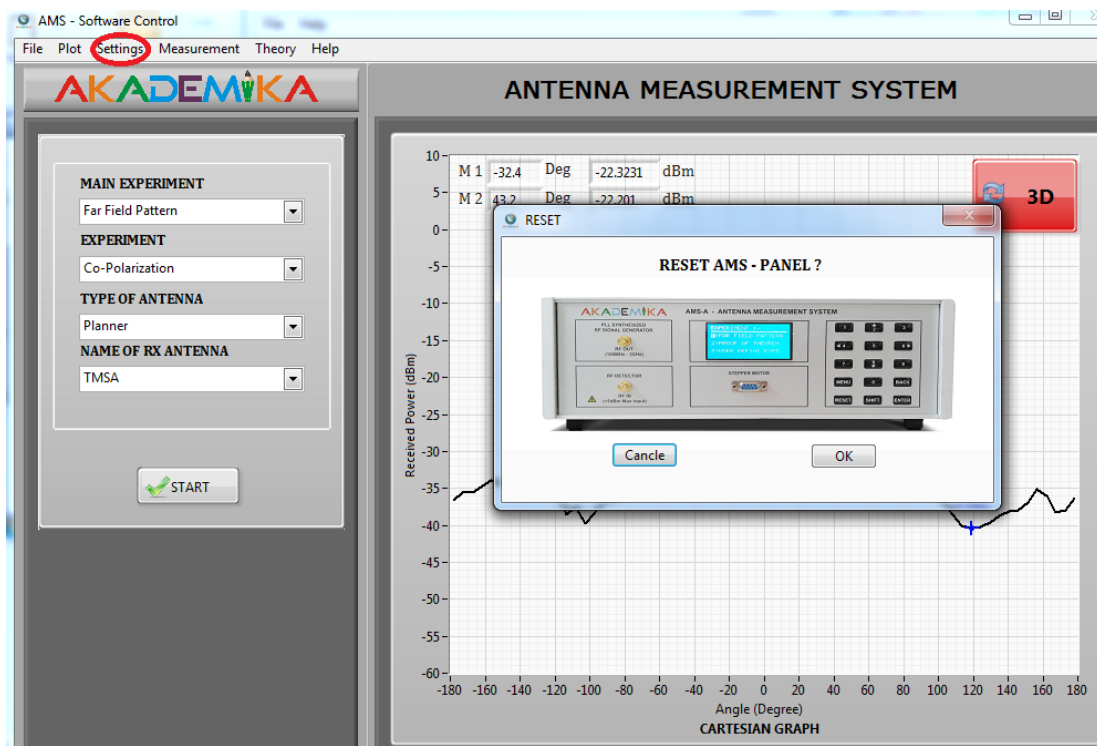
9. You can switch over the Polar Plot to Cartesian Graph from MENU:
Plot: Cartesian Graph and can save the graph through **File: Save: Cartesian Graph.**



10. For Measurements, Click on Menu option **Measurement**. A tab will be opened at left corner. Select Type and Name of Receiver Antenna and click on the List of various Antenna factors provided, and get values.

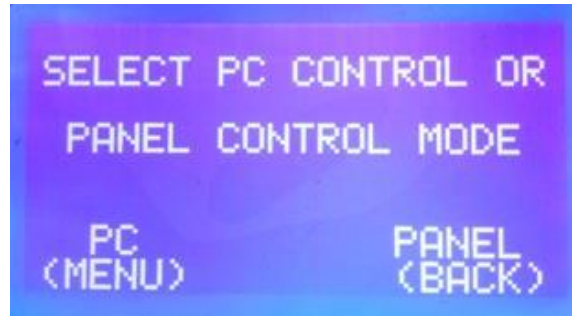


11. You can RESET the AMS trainer Kit for changing the control mode through MENU option **Reset AMS**. Then close the software and re-open to change the control mode.

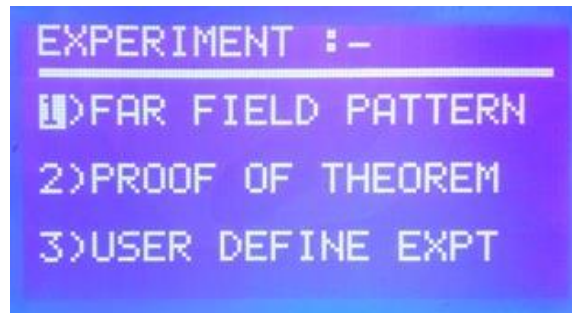


B. PANEL CONTROL MODE

1. Select Control Mode as PANEL by Pressing **BACK** button on PANEL.



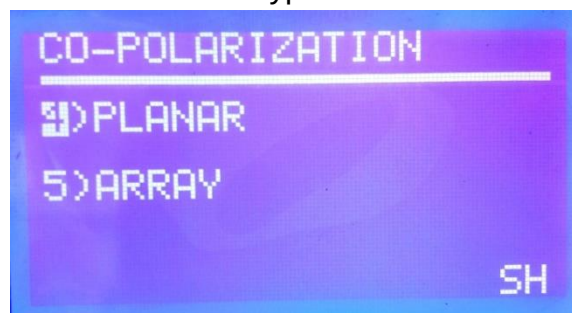
2. Select Main Experiment as **Far Field Pattern**.



3. Select **Co-Polarization** as Experiment from the panel.



4. Use **SHIFT** button to Scroll. Select Type of Antenna as **PLANNER**.



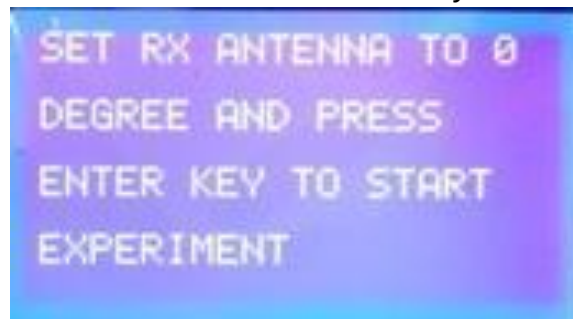
5. Select Antenna as **TMSA**.



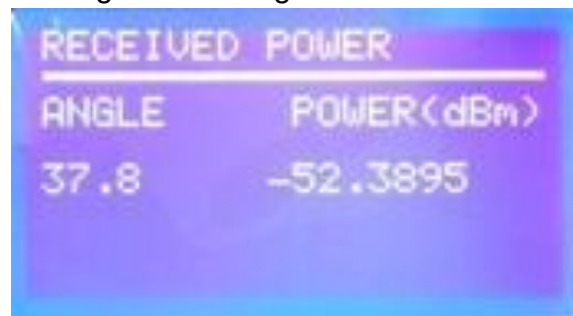
6. Information of selected antenna will be shown as below;



7. Set Receiver Antenna to 0° and Press ENTER key to start the experiment.



8. Antenna will start rotating and reading will be taken.

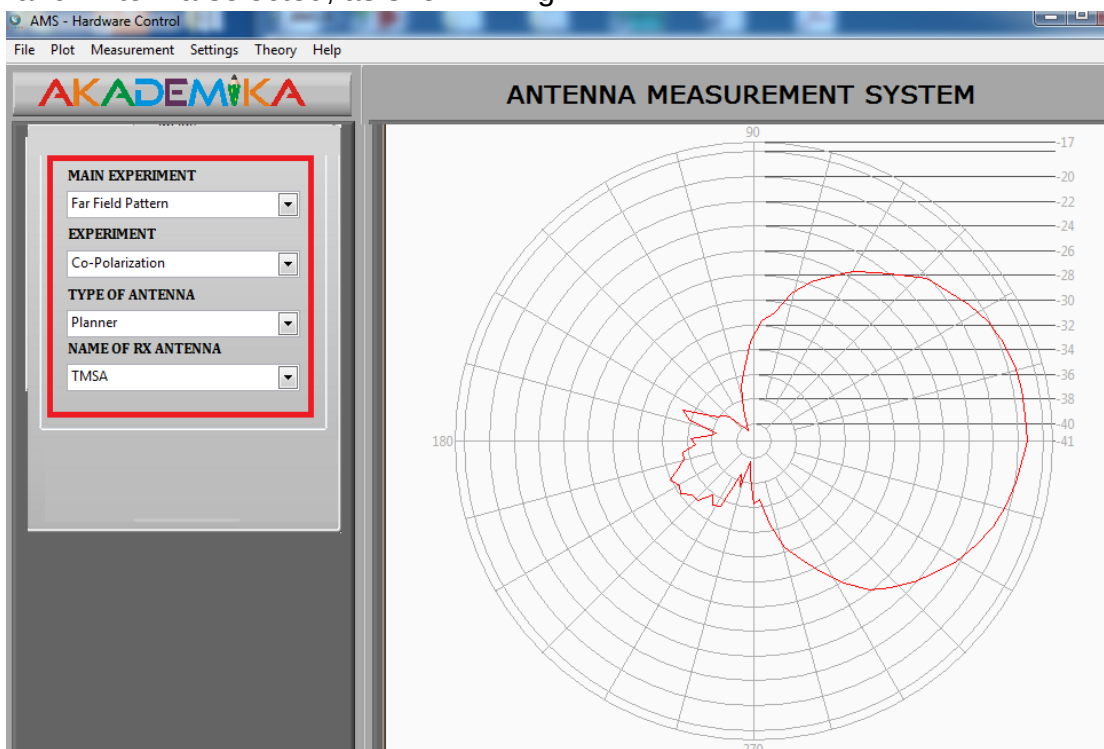


9. After completion of all the 68 readings, it will ask to upload the Readings on Computer. Confirm the selection using ENTER button. Make sure the software is Running and press ENTER again.

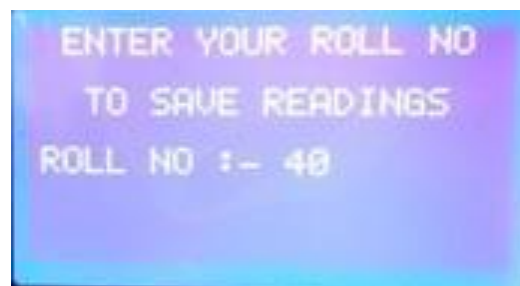
UPLOAD DATA
TO COMPUTER?????

MAKE SURE THAT,
SOFTWARE IS RUNNING

10. It will then plot the graph on Software screen PULLING UP the Experiment details and Antenna selected, as shown in fig.



11. After completion of the readings it will ask to **Save the Readings**, press **ENTER** to save the readings. Enter Roll number where the readings will be saved (Roll no. Limit is 60).



12. To **Open the Readings**, Select **PANEL** control from **AMS Trainer kit** and Select **RECALL READINGS** from experiment.



13. Enter **Roll No.** and Press **ENTER**.



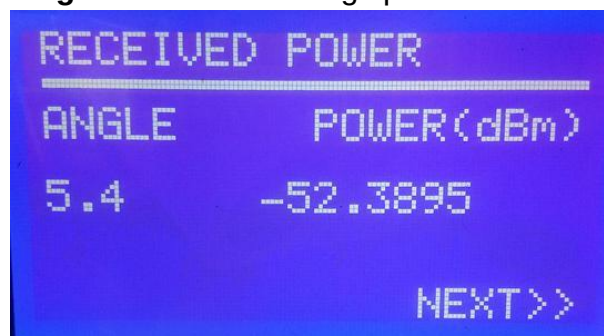
14. Available list of Experiments will be shown. Select any Experiment and Press ENTER.



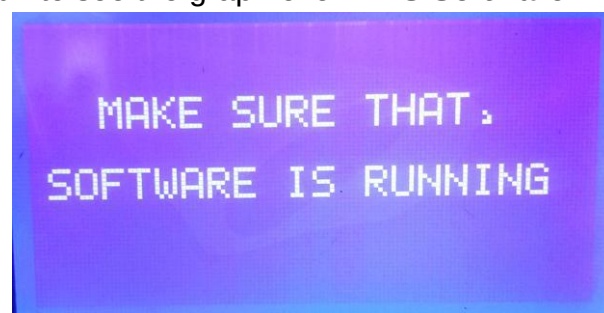
15. Two options will be there :
VIEW READINGS & PLOT GRAPH.



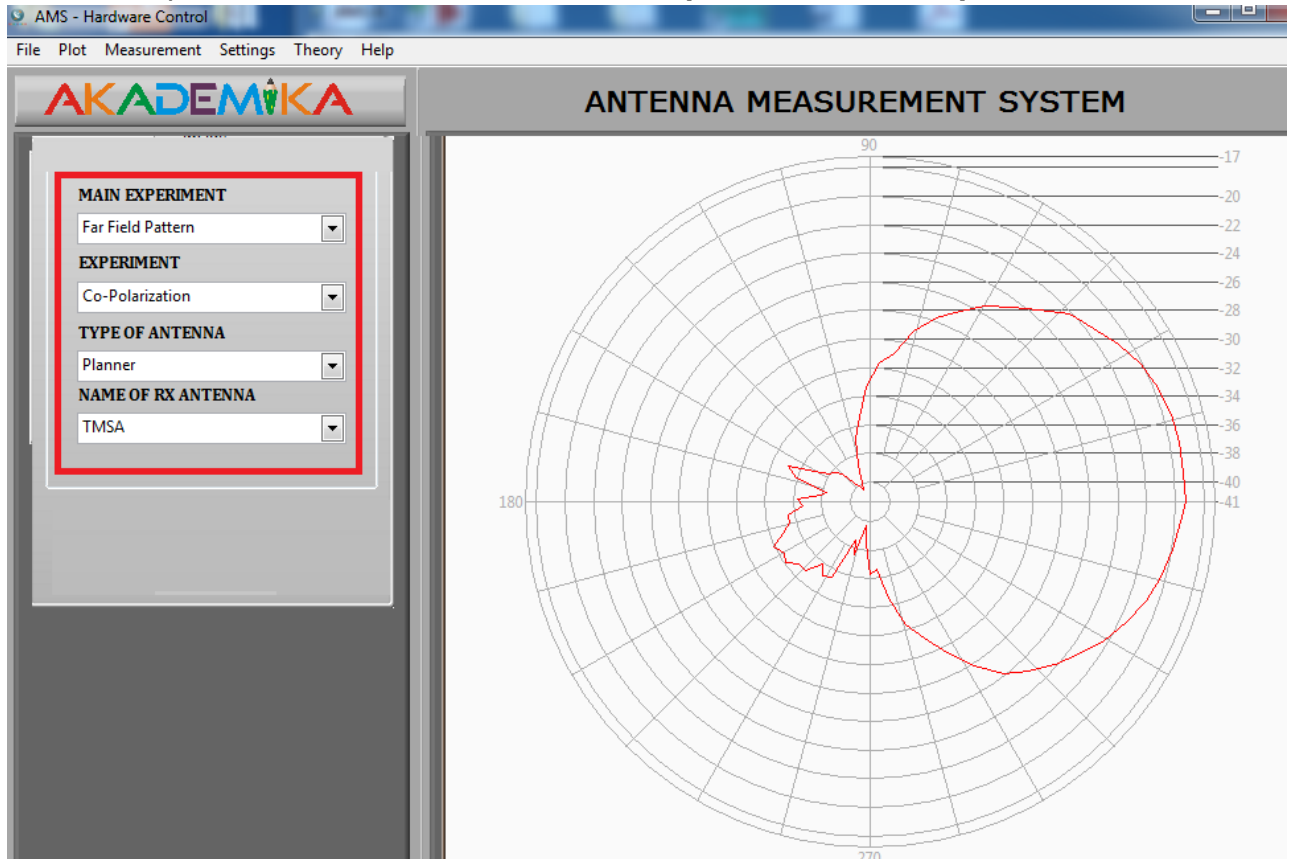
16. Select **View Readings** to see the readings present at that roll no.



17. Select **Plot Graph** to see the graph over **AMS Software**.



18. The plot will be drawn at **AMS Software (Hardware Control)**.



19. At Software part (Hardware control), again the **Save Readings, Save Image, Open Readings, Plot Selection (Polar, Cartesian) & Measurements** selection is same as the **Software Control** mode discussed earlier.

20. There are again more Experiments such as,
- **Proof Of Theorems**
 - **Reciprocity**
 - **Inverse Square Law**
 - **User Defined Experiments**
 - **RSSI (Received signal strength Indicator)** etc.
- They all can be achieved using above procedure.

EXPERIMENT NO. 1

EXPERIMENT NO : 1

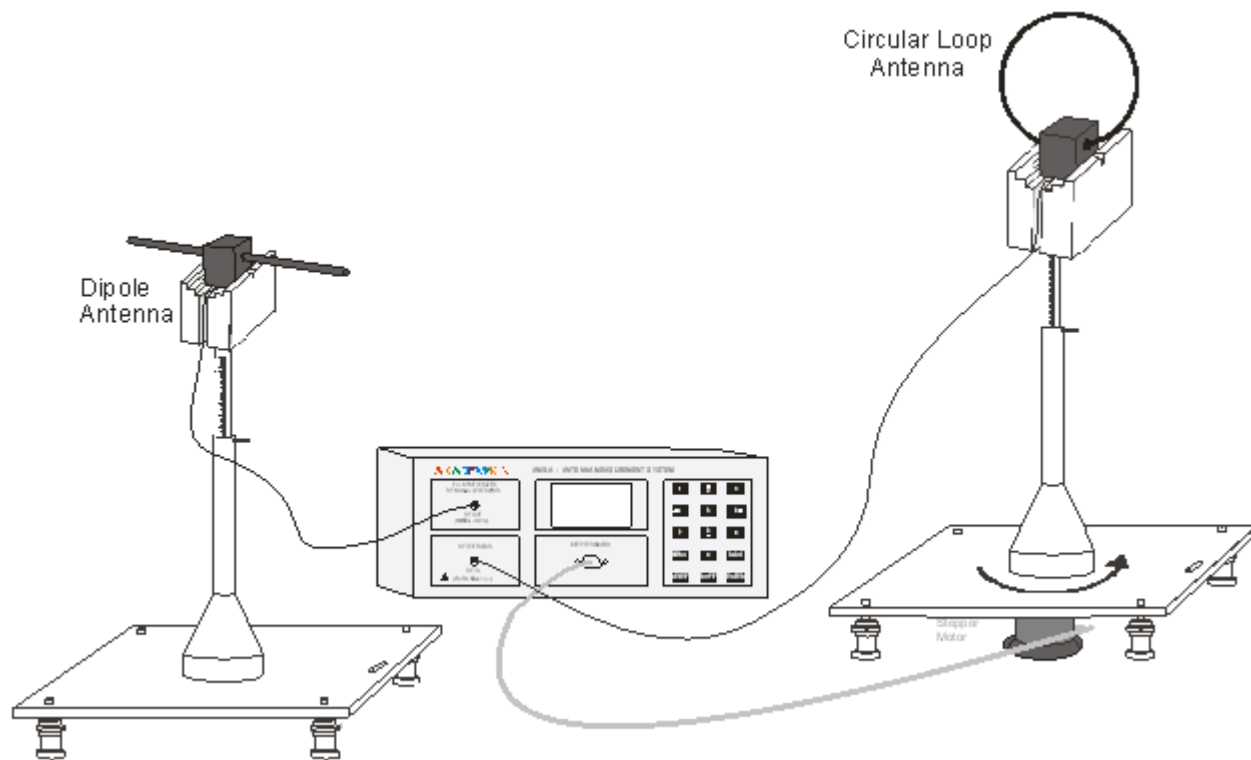
OBJECTIVE

To Measure the variation of field strength of radiated wave, with distance from transmitting antenna.

EQUIPMENT REQUIRED:

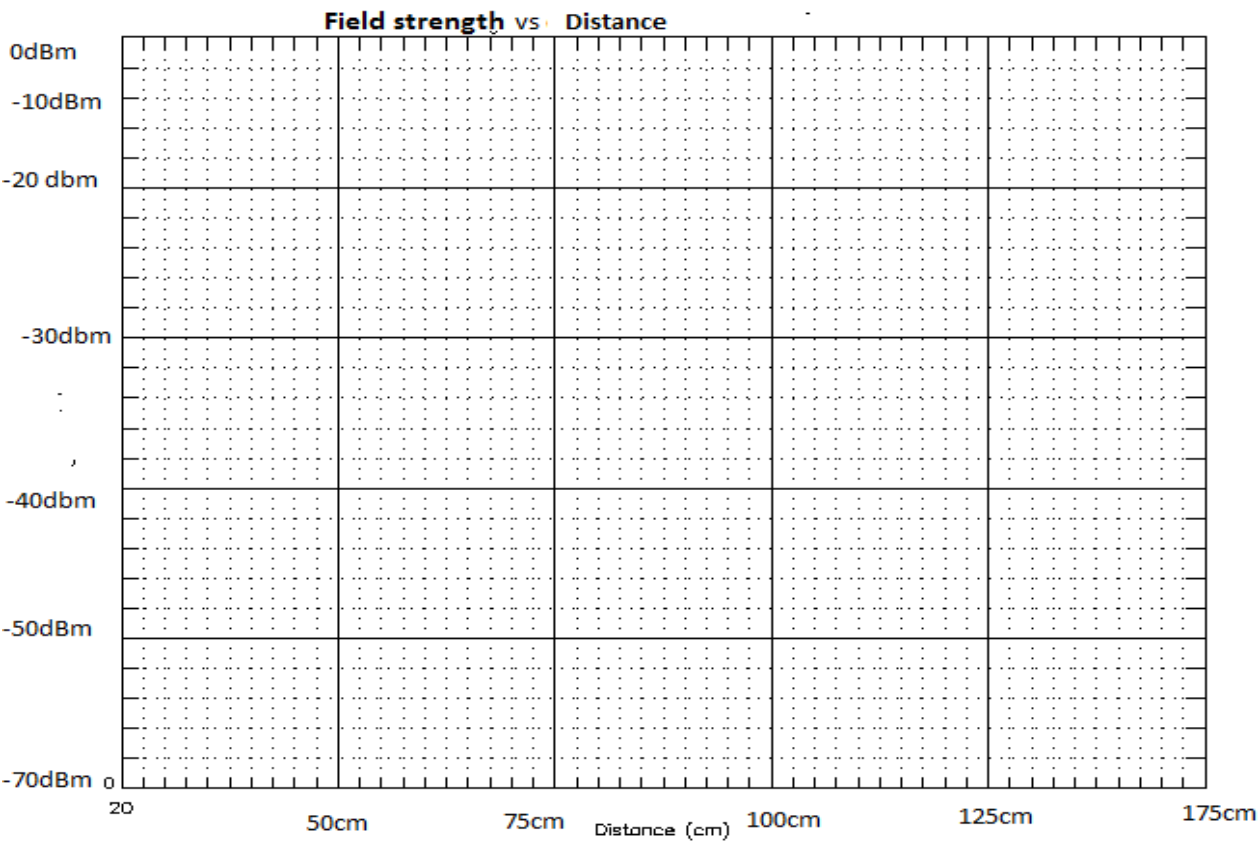
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire or Microstrip DIPOLE	Any antenna	Source (RF output)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Turn ON the module, select control mode.
3. Open the AMS-A.exe file Select the corresponding COM PORT and Click on Run, now the software will be in running mode.
4. Select the PROOF OF THEOREM and then select the experiment of INVERSE SQUARE LAW.
5. Then select the RX antenna and click on START button.
6. A Pop-Up window will appear asking user to Keep the Distance between TX and RX as 10cm then click on NEXT button.
7. After completion of one reading again a popup window will appear asking to Increase the TX and RX distance by 10cm.
8. Repeat the procedure up to a distance of 100cm at an interval of 10cm; click On NEXT to plot the graph.
9. After that it will plot the radiation pattern in inverse square law graph (i.e. Distance in cm V/S power in dBm).
10. While taking the readings, ensure that no scattering objects are in the vicinity of the antenna, this could reradiate and distort the field pattern and consequently the readings. Avoid any movement of persons while taking the readings.
11. Plot these readings manually on graph paper with distance between Antennas on X axis and signal level in dB at Y-axis.
12. Use the graph template provided below for plotting your graph.
13. Now take the readings and observe the plots for different types of antenna.



EXPERIMENT NO. 2

EXPERIMENT NO. 2

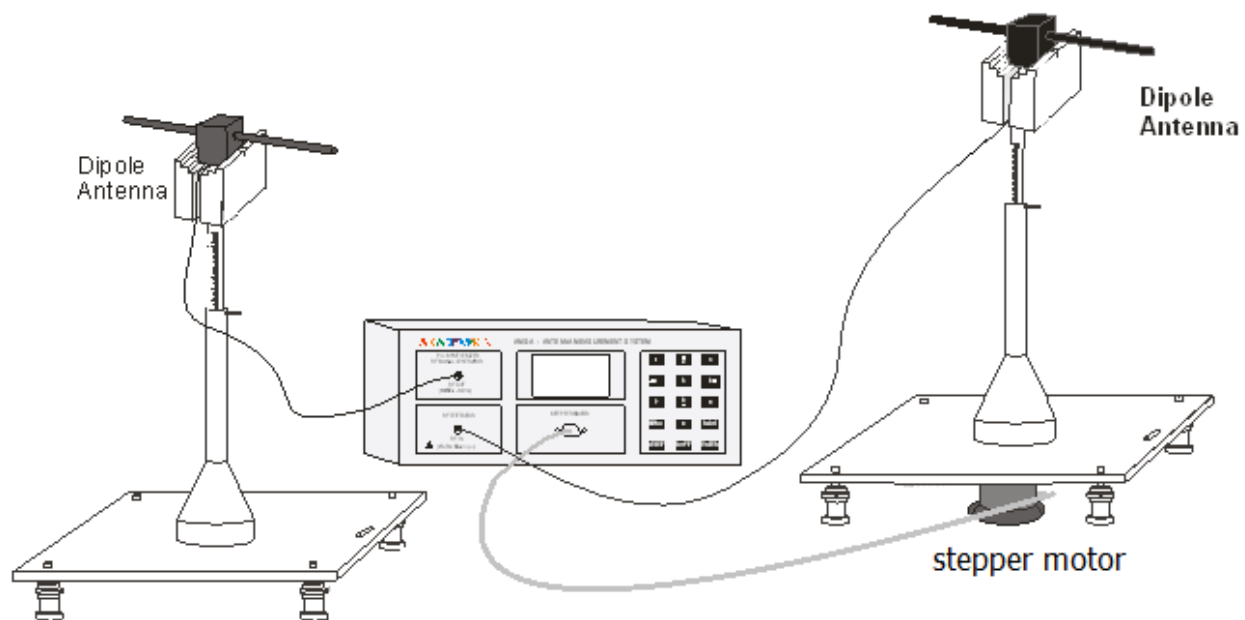
OBJECTIVE

To demonstrate that the transmitting and receiving radiation patterns of an antenna are equal and hence confirm the reciprocity theorem of antennas.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Wire Dipole	Source (RF out)	Detector (RF input)	SMA To SMA Cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details of far field region.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file Select the corresponding COM PORT and Click on Run, now the software will be in running mode.
5. Select the **PROOF OF THEOREM** and then select the experiment of **RECIPROCITY THEOREM**.
6. Then select the RX antenna and click on START button.
7. After completing it will plot the readings.
8. Now interchange the antenna and again take the next reading.
9. Compare both the plots they must be same, hence reciprocity theorem is Proved.
10. Similarly do the experiments for different types of antenna.

NOTE:

1. **Keep in mind that an antenna that is being rotated is plotted in reception and transmission mode both for proving the reciprocity theorem.**
2. **Observe the two plots and they must be approximately same.**

EXPERIMENT NO. 3

EXPERIMENT NO. 3

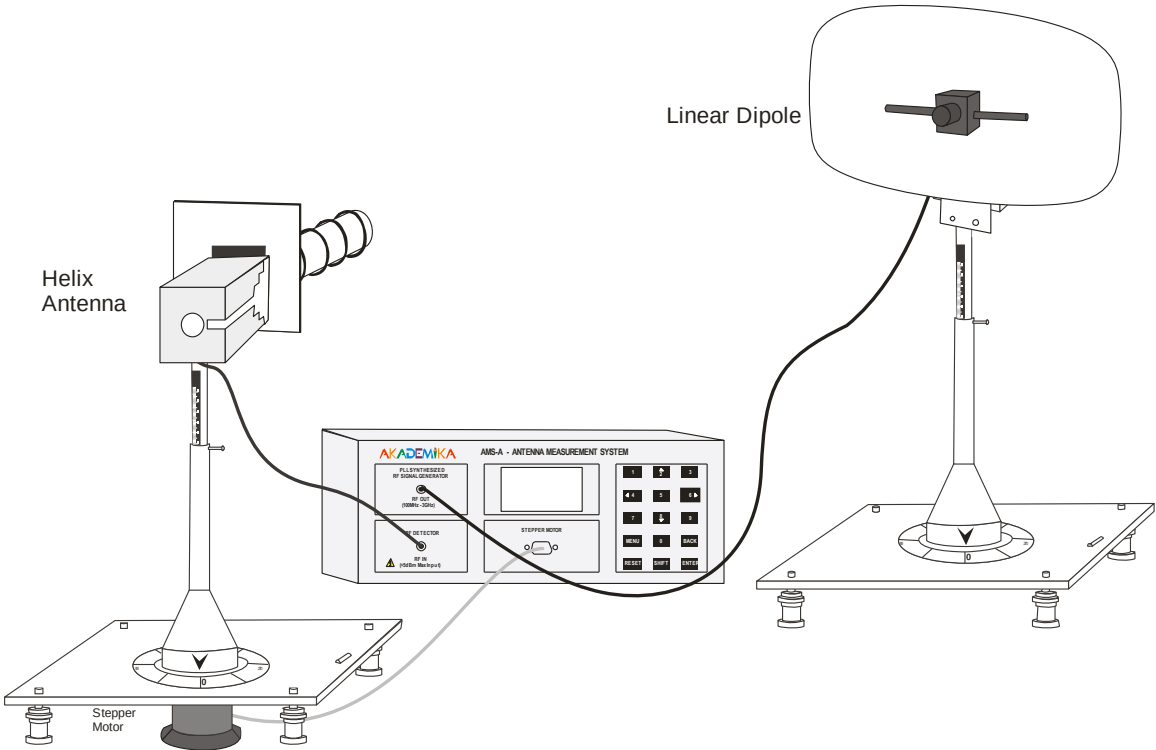
OBJECTIVE

To plot the Radiation pattern of all WIRED antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire DIPOLE	Any wired antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

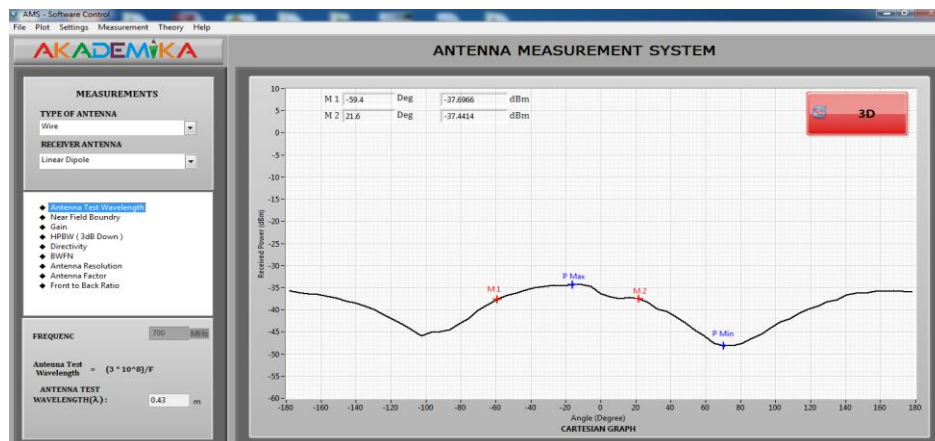
SET UP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **NULL & SIDELobe**, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.

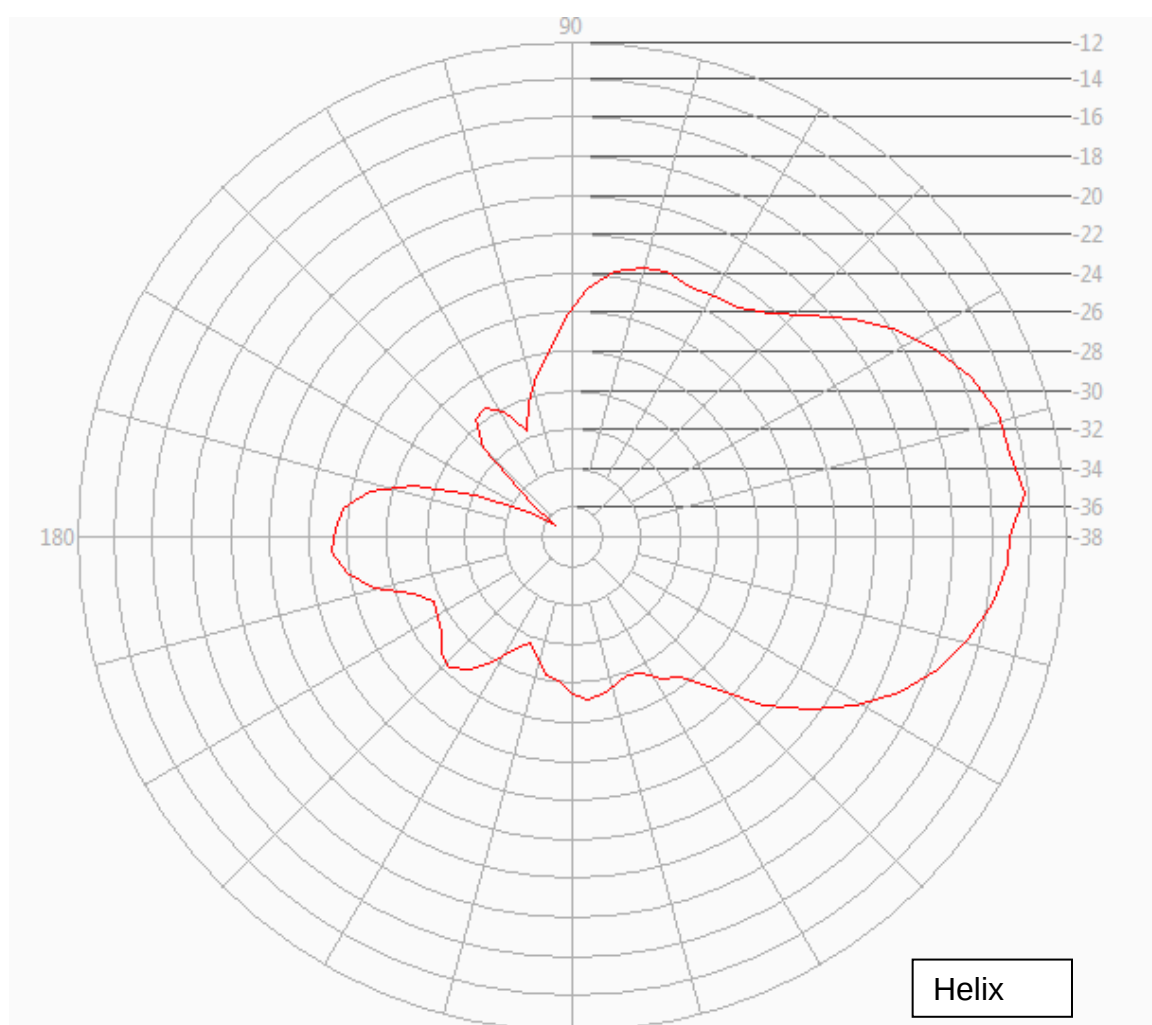
(For the formulas used to calculate the above parameters refer Annexure-1)



8. Now repeat procedure for other wire antenna and observe the plot.
9. From SAVE option, the plot can be saved and take the printout whenever required.

Example 1)

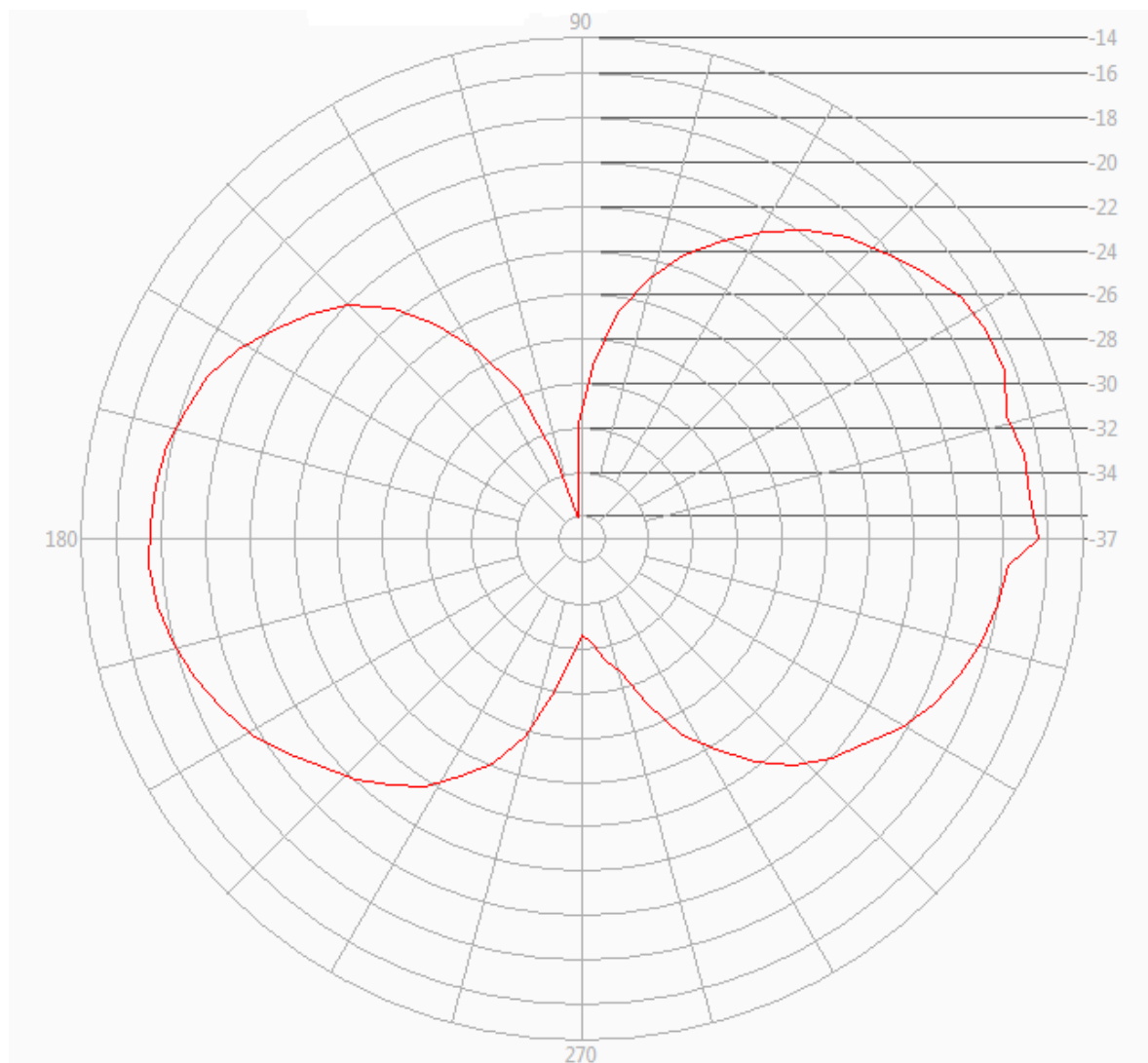
TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Helix	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

Example 2)

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Folded Dipole	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

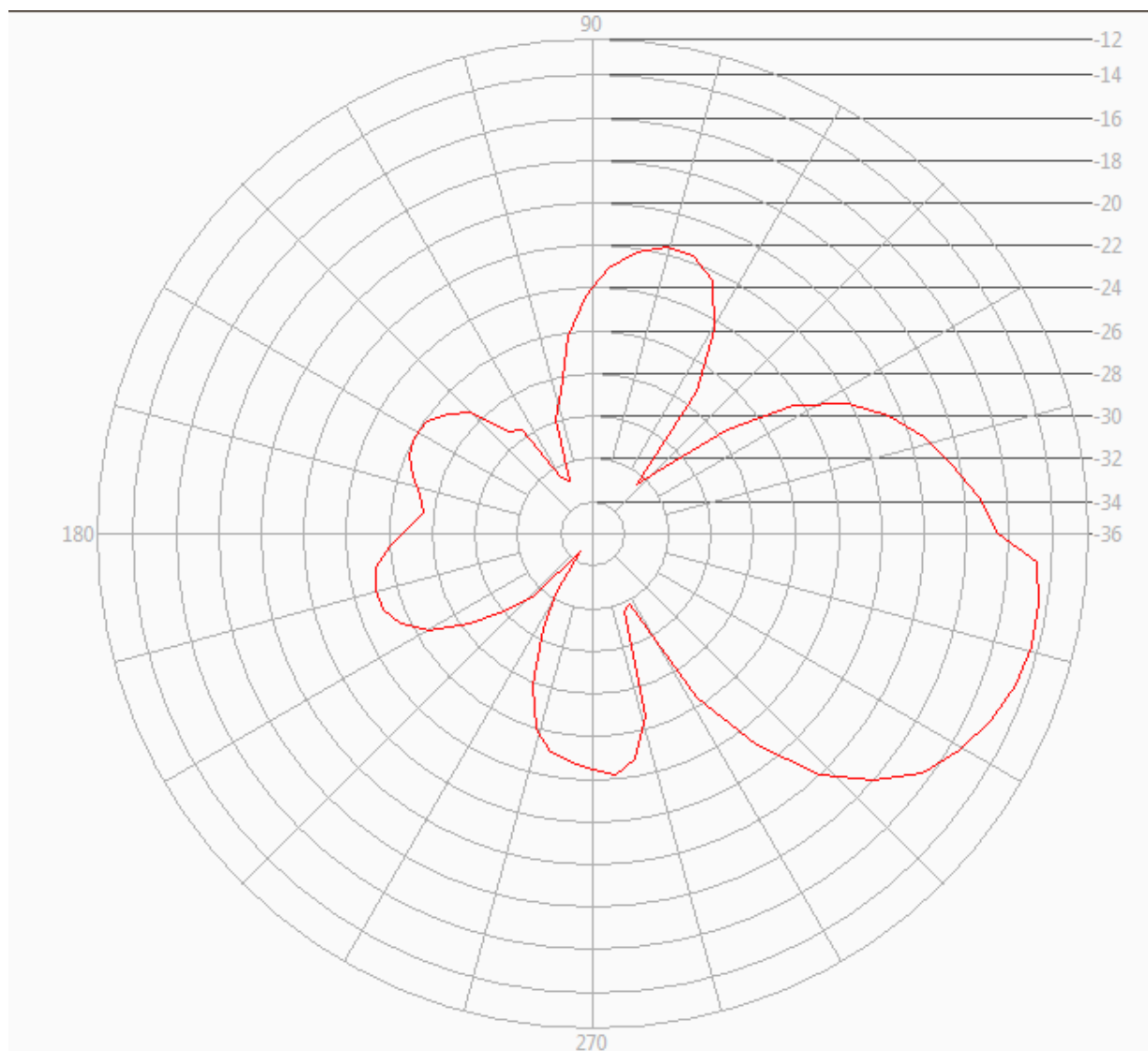
- RADIATION PATTERN:**



Example 3)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Yagi Uda (5 elements)	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

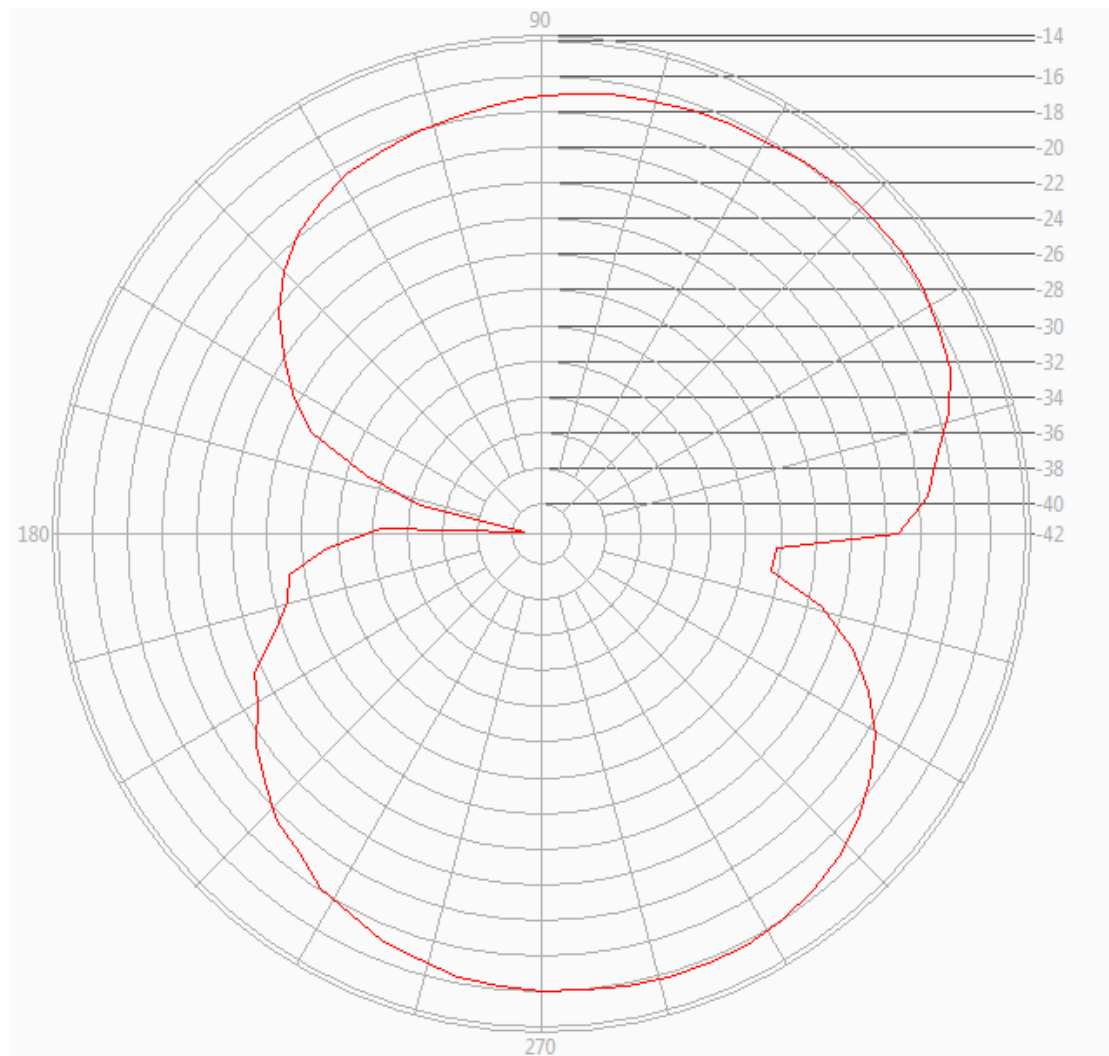
- RADIATION PATTERN:**



Example 4)

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Wire monopole	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

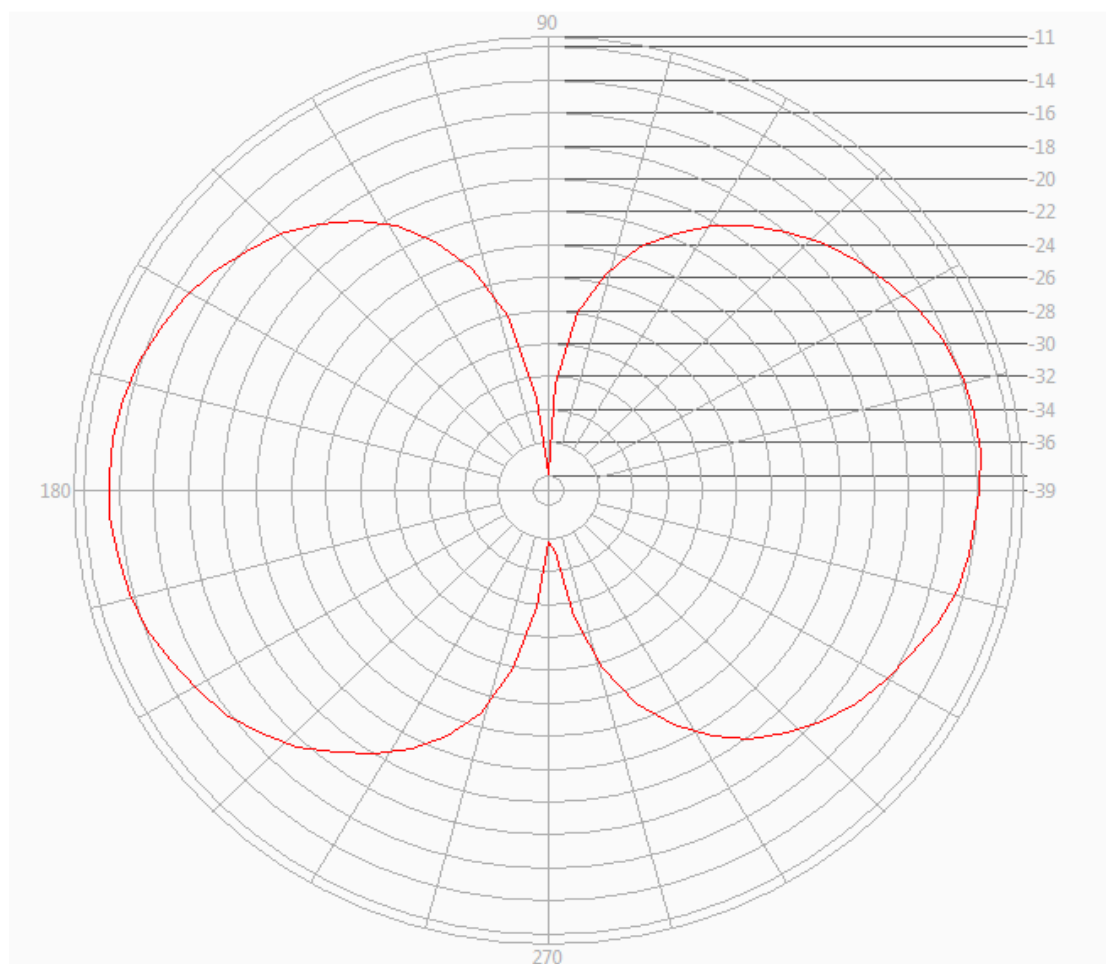
- RADIATION PATTERN:**



Example 5)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Rectangular loop	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

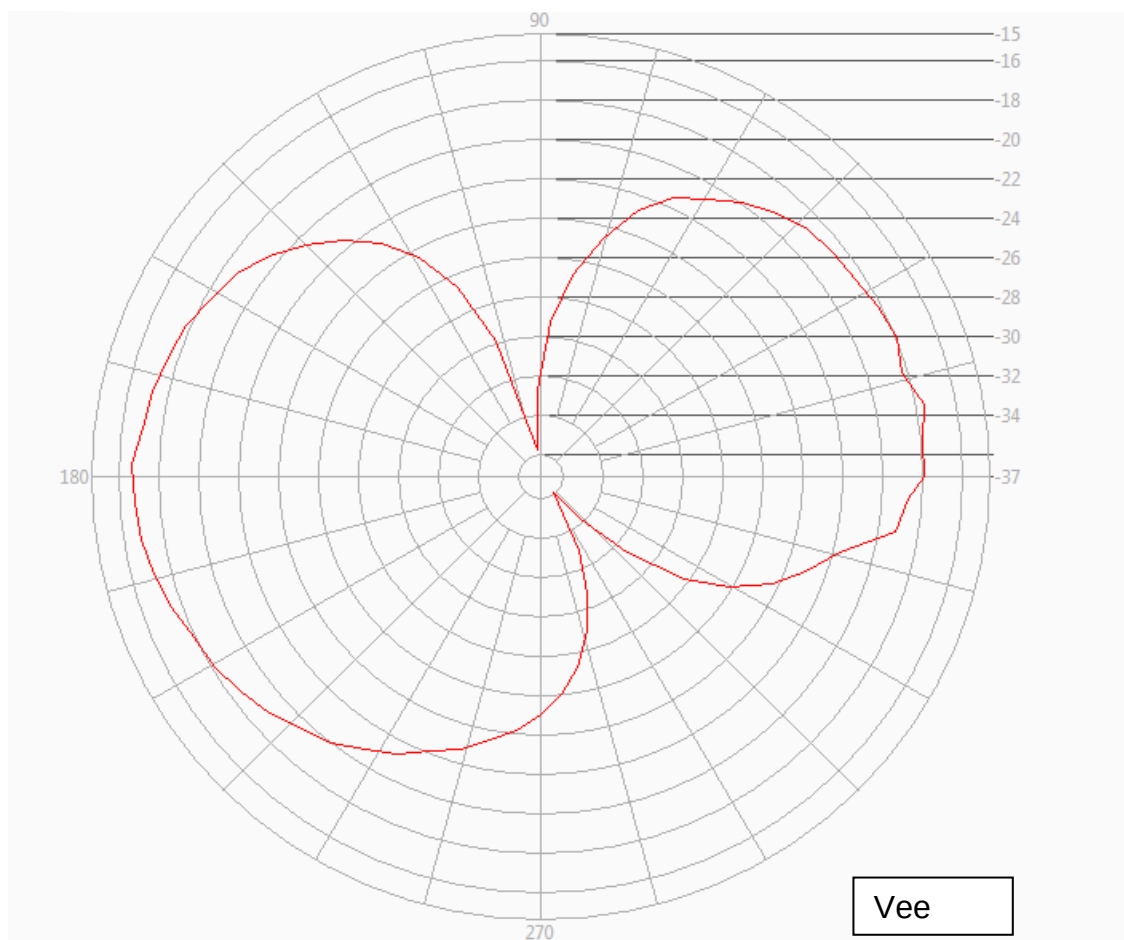
- RADIATION PATTERN:**



Example 6)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	VEE	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

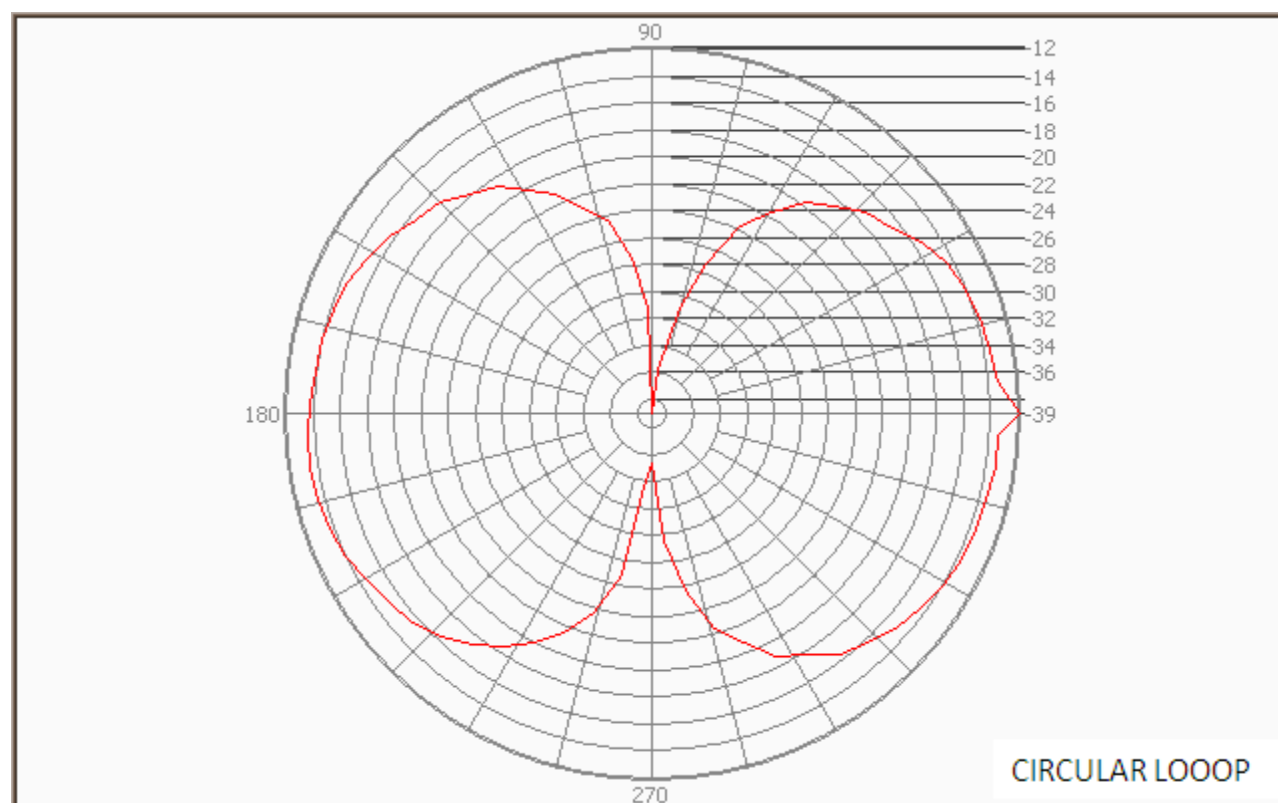
- RADIATION PATTERN**



Example 7)

TX antenna	RX antenna	Transmitter input	Receiver output	Cable
Wire Dipole	Circular loop	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

- RADIATION PATTERN:**



EXPERIMENT NO. 4

EXPERIMENT NO. 4

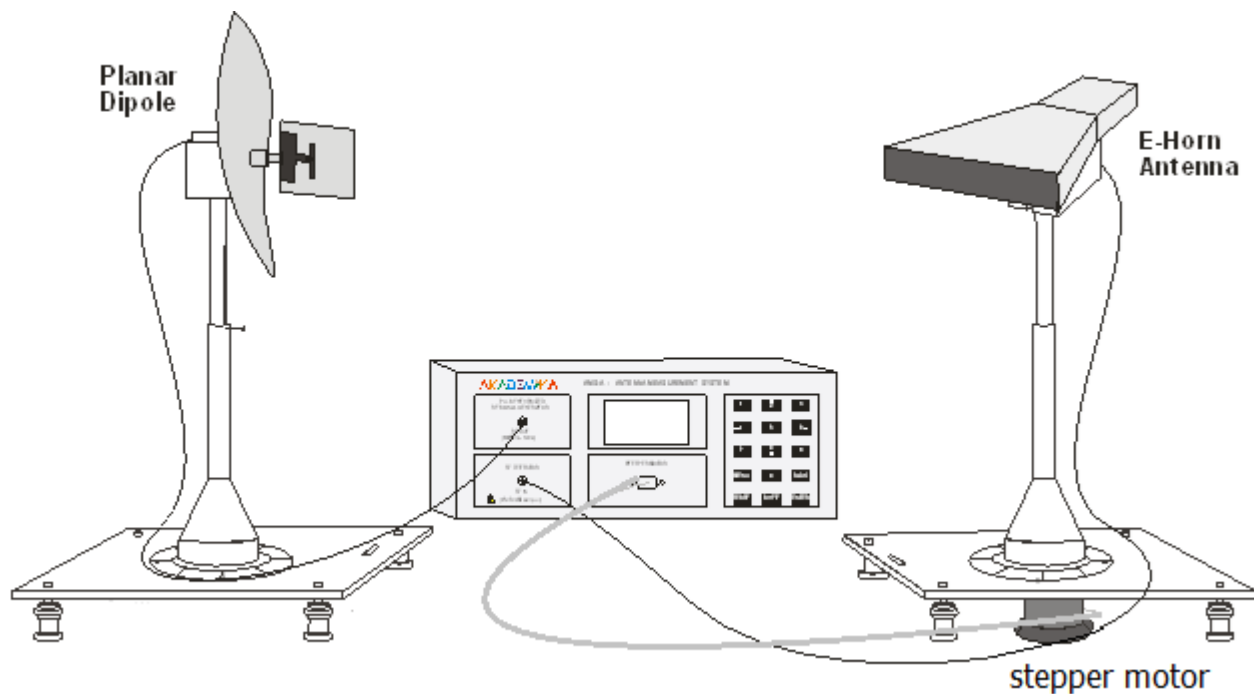
OBJECTIVE

To plot the radiation pattern of all APERTURE antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	Cable
Planar DIPOLE	Any APERTURE antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SET UP ARRANGEMENT:



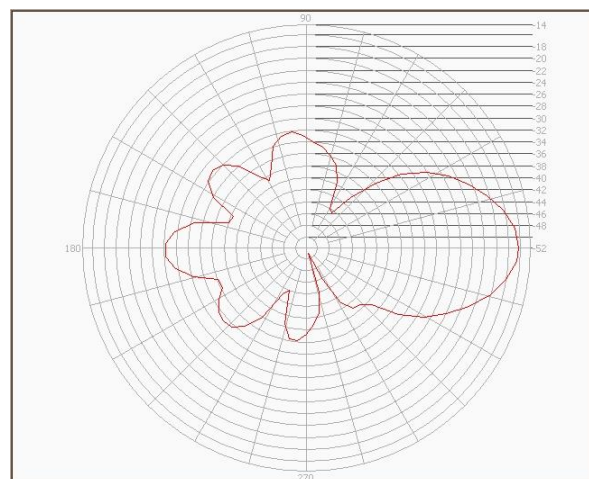
PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **CO-POLARIZATION**, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.
(For the formulas used to calculate the above parameters refer Annexure-1)
8. From SAVE option, the plot can be saved and take the printout whenever required.

Example

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	E-Horn	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

- RADIATION PATTERN**



EXPERIMENT NO. 5

EXPERIMENT NO. 5

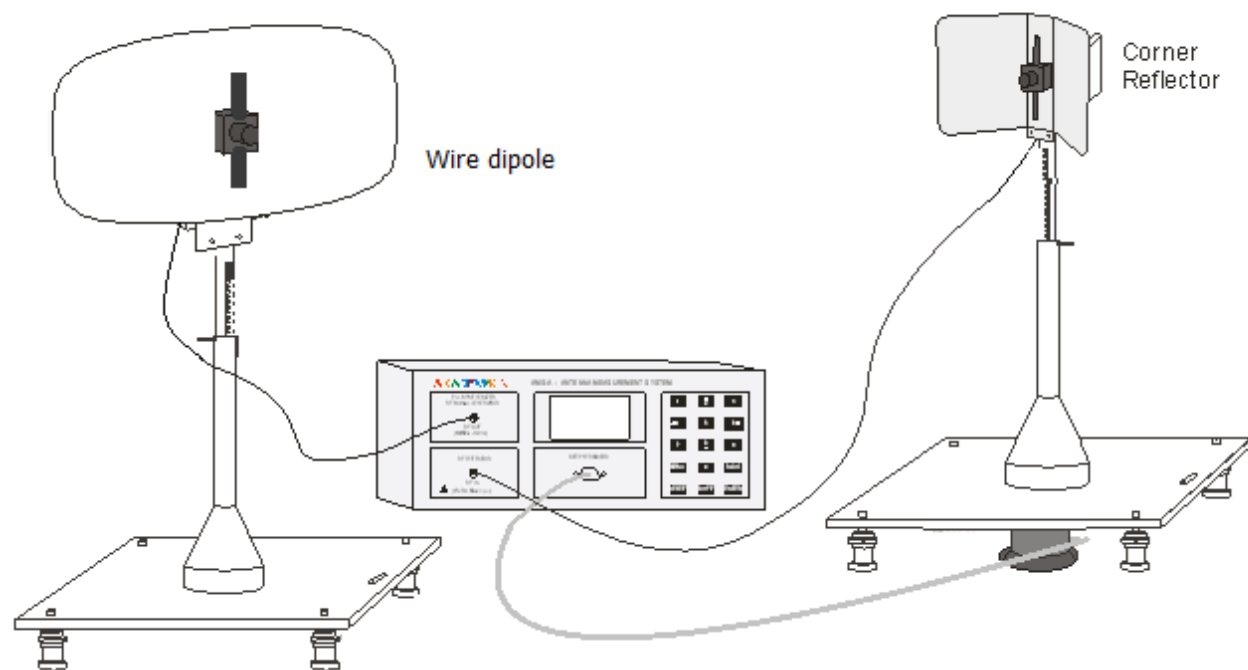
OBJECTIVE

To plot the radiation pattern of all REFLECTOR antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire DIPOLE	Corner Reflector antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

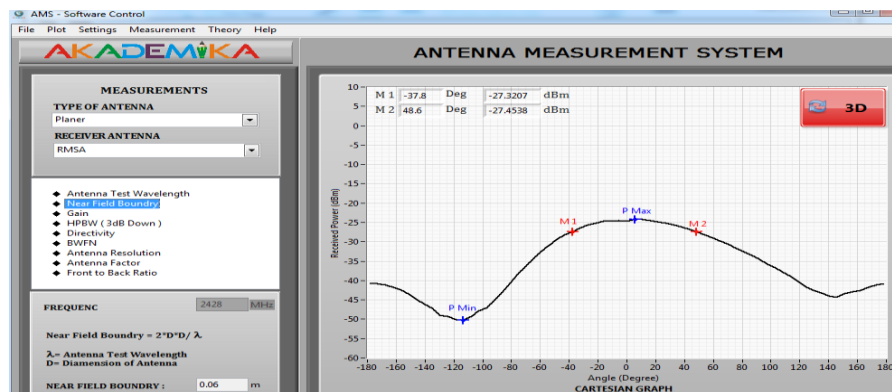
SET UP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **Gain / Directivity**, select **Reflector** and select RX antenna as corner then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.

Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.



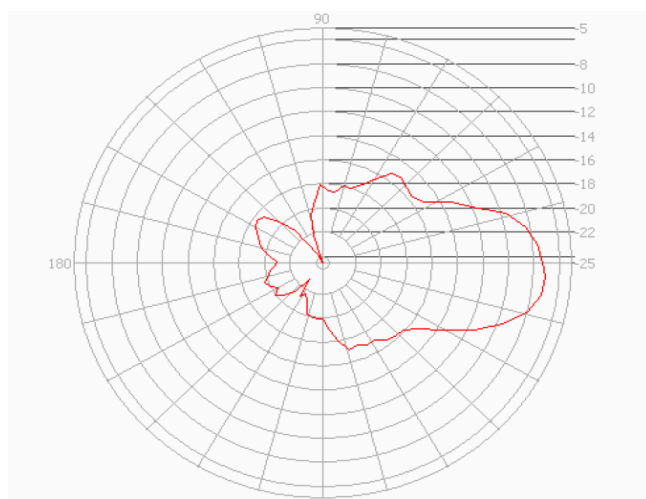
(For the formulas used to calculate the above parameters refer Annexure-1)

7. From SAVE option, the plot can be saved and take the printout whenever required.

Example

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Corner reflector	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

- **RADIATION PATTERN:**



EXPERIMENT NO. 6

EXPERIMENT NO. 6

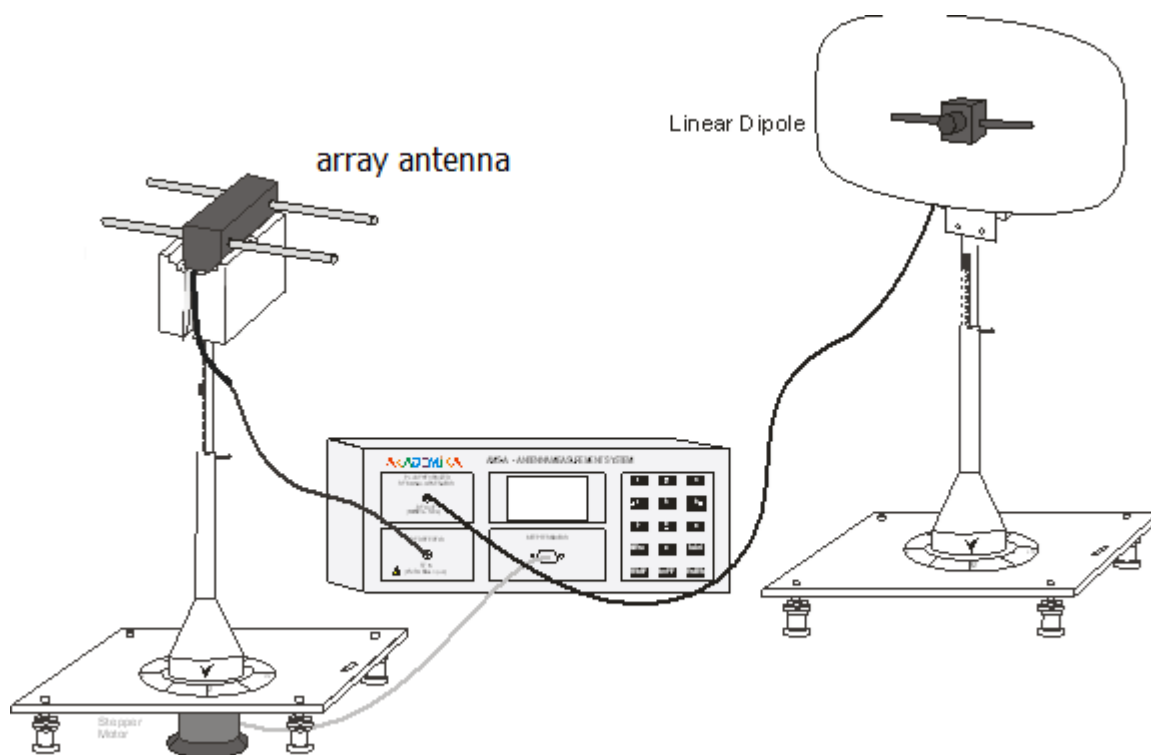
OBJECTIVE

To plot the radiation pattern of all ARRAY antenna and observe its parameters.

EQUIPMENT REQUIRED:

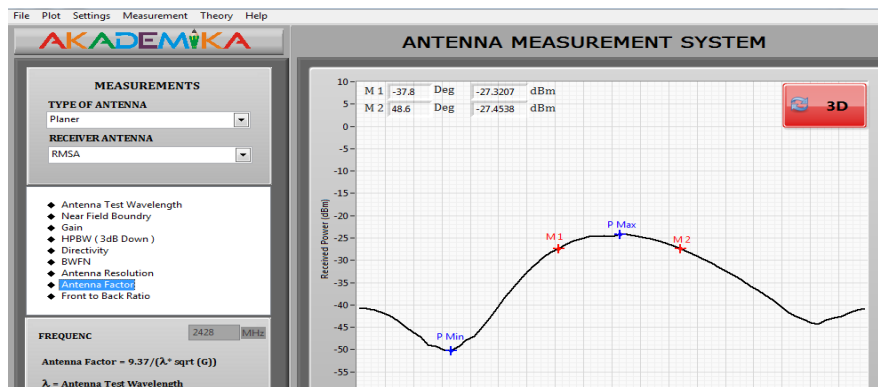
TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire DIPOLE	Any ARRAY antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SET UP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 1meter, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **NULL& SIDELobe**, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.



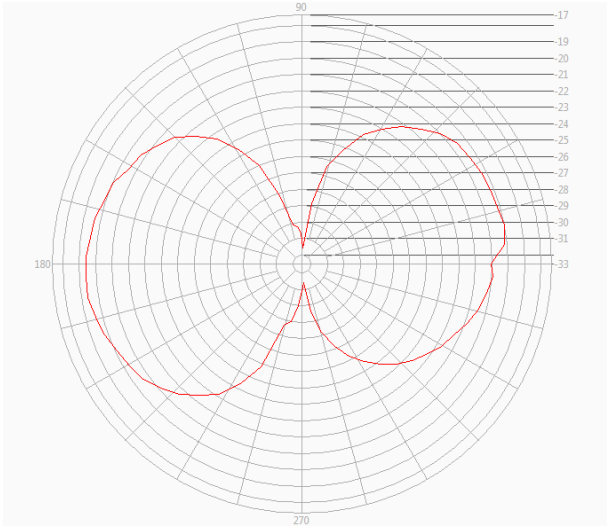
(For the formulas used to calculate the above parameters refer Annexure-1)

8. From SAVE option, the plot can be saved and take the printout whenever required.

Example 1)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Broadside array	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

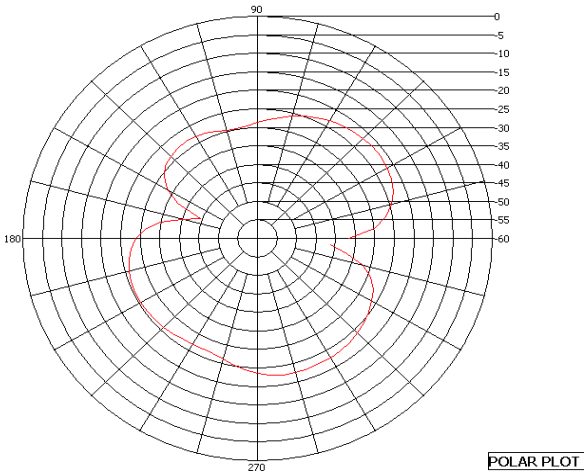
RADIATION PATTERN:



EXAMPLE 2)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	End fire	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:



EXPERIMENT NO. 7

EXPERIMENT NO. 7

OBJECTIVE

To observe the effect of CO-POLARIZATION and CROSS-POLARIZATION using vertical and horizontal polarization of antenna.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
1. WIRE DIPOLE 2. PLANAR(Microstrip) DIPOLE	1. Any WIRE antenna 2. Any MICROSTRIP antenna	Source (RF out)	Detector (RF input)	To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

PROCEDURE:

CO-POLARIZATION:



1. Set up the experiment as per shown in figure above.
2. Set transmitting as well as receiving antenna in same polarization i.e. VERTICAL/HORIZONTAL.
3. Set the distance between the antennas to be around 1m, consult theory for details of far field region.
4. Turn ON the module, select control mode.
5. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
6. Go in **FAR FIELD PATTERN**, select **CO-POLARIZATION**, select RX (either WIRE or PLANAR) antenna then click on START
7. Then readings from 0 degree to 360 degree will be plotted in the software.
8. After completing it will plot the readings.

CROSS POLARIZATION:

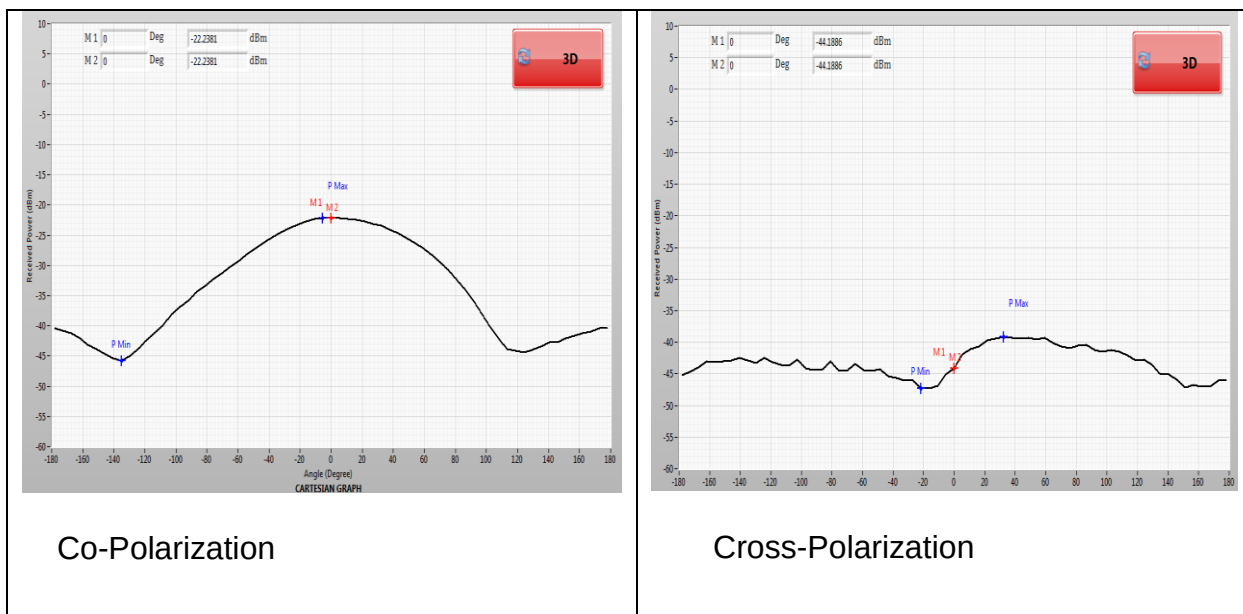


1. Set up the experiment as per shown in figure above.
2. Set transmitting as well as receiving antenna in opposite/different polarization i.e. VERTICAL/HORIZONTAL.

3. Set the distance between the antennas to be around 1meter, consult theory for details of far field region.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **CROSS-POLARIZATION**, select RX (either WIRE or PLANAR) antenna then click on START.
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. After completing it will plot the readings.
8. Observe the effect of change in polarization of antenna.

EXAMPLE:

TX antenna	RX antenna	TX frequency	Separation	Measurement
Planar Dipole	TMSA	2437 MHz	60cm	Co polarization Cross-Polarization



EXPERIMENT NO. 8

EXPERIMENT NO. 8

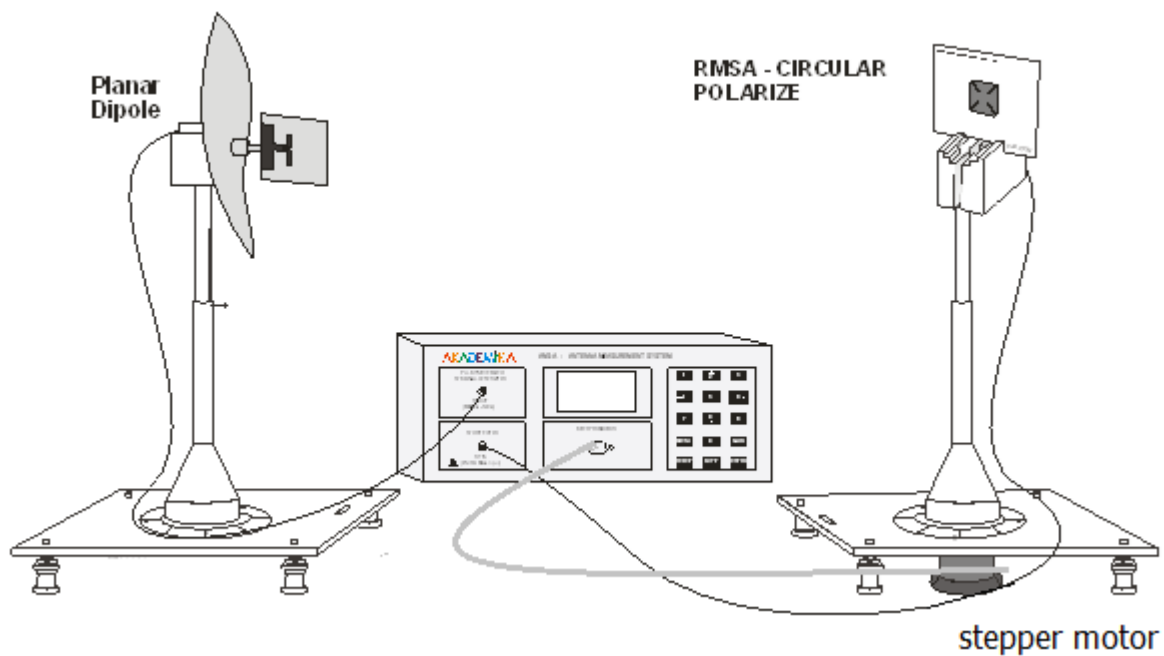
OBJECTIVE

Measurement of Circularly polarized antennas and observe its parameters.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
1. WIRE DIPOLE 2. PLANAR (Microstrip) DIPOLE	1. Helix 2. Cross Dipole 3. RMSA-Circular Polarized	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SET UP ARRANGEMENT:



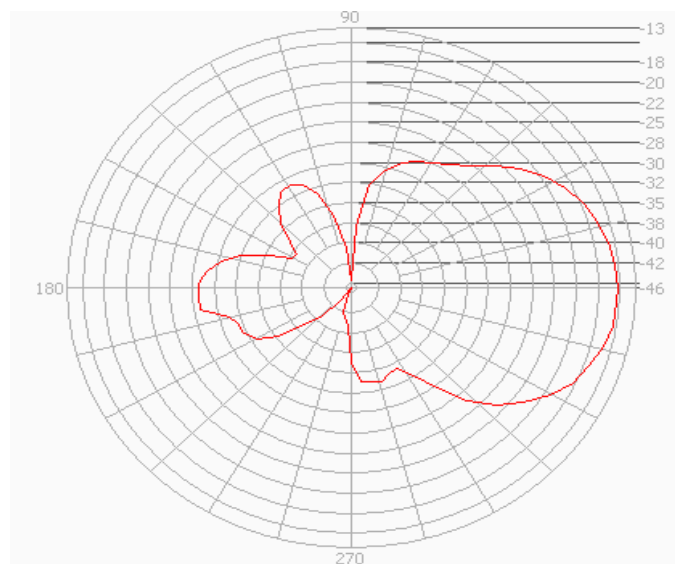
PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **Front-Back Ratio**, select **Wire**, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Thus the circular polarized antenna receives a maximum power in both the polarization.
8. From SAVE option, the plot can be saved and take the printout whenever required.

Example 1)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Wire Dipole	Helix	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

- RADIATION PATTERN:**



EXPERIMENT NO. 9

EXPERIMENT NO. 9.1

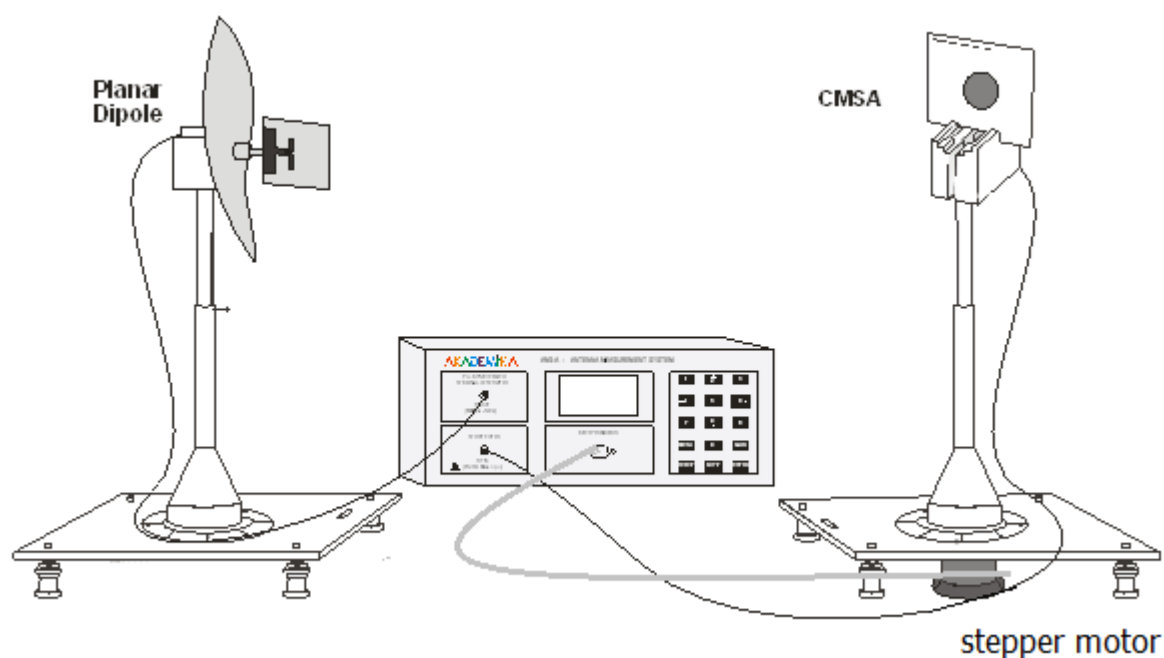
OBJECTIVE:

To plot the radiation pattern of all PLANAR (MICROSTRIP) antenna and observe its parameters.

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	Any PLANAR antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

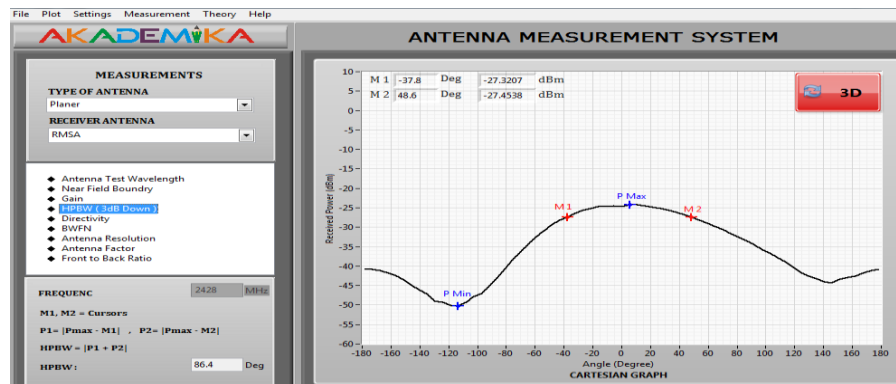
SET UP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **CO-POLARIZATION**, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.

(For the formulas used to calculate the above parameters refer Annexure-1)

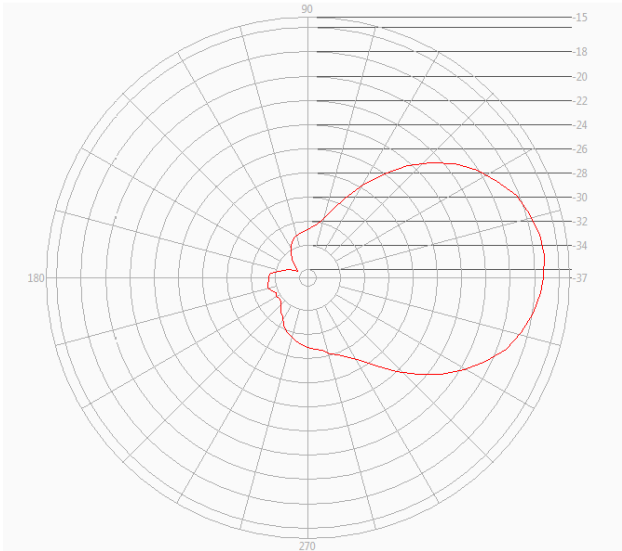


8. Now repeat for CROSS-POLARIZATION of antenna and observe the plot.
9. From SAVE option, the plot can be saved and take the printout whenever required.

Example 1)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planar Dipole	2 X 1 Array	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN:

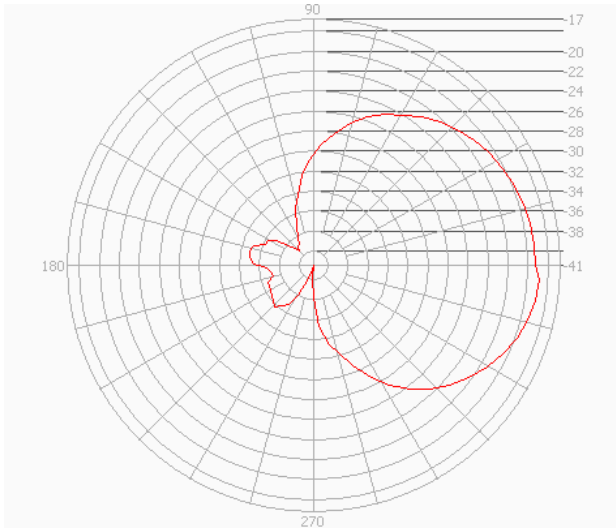


For field pattern – Front – Back Ratio – planer - Insert feed.

EXAMPLE 2)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planar Dipole	Insert feed	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

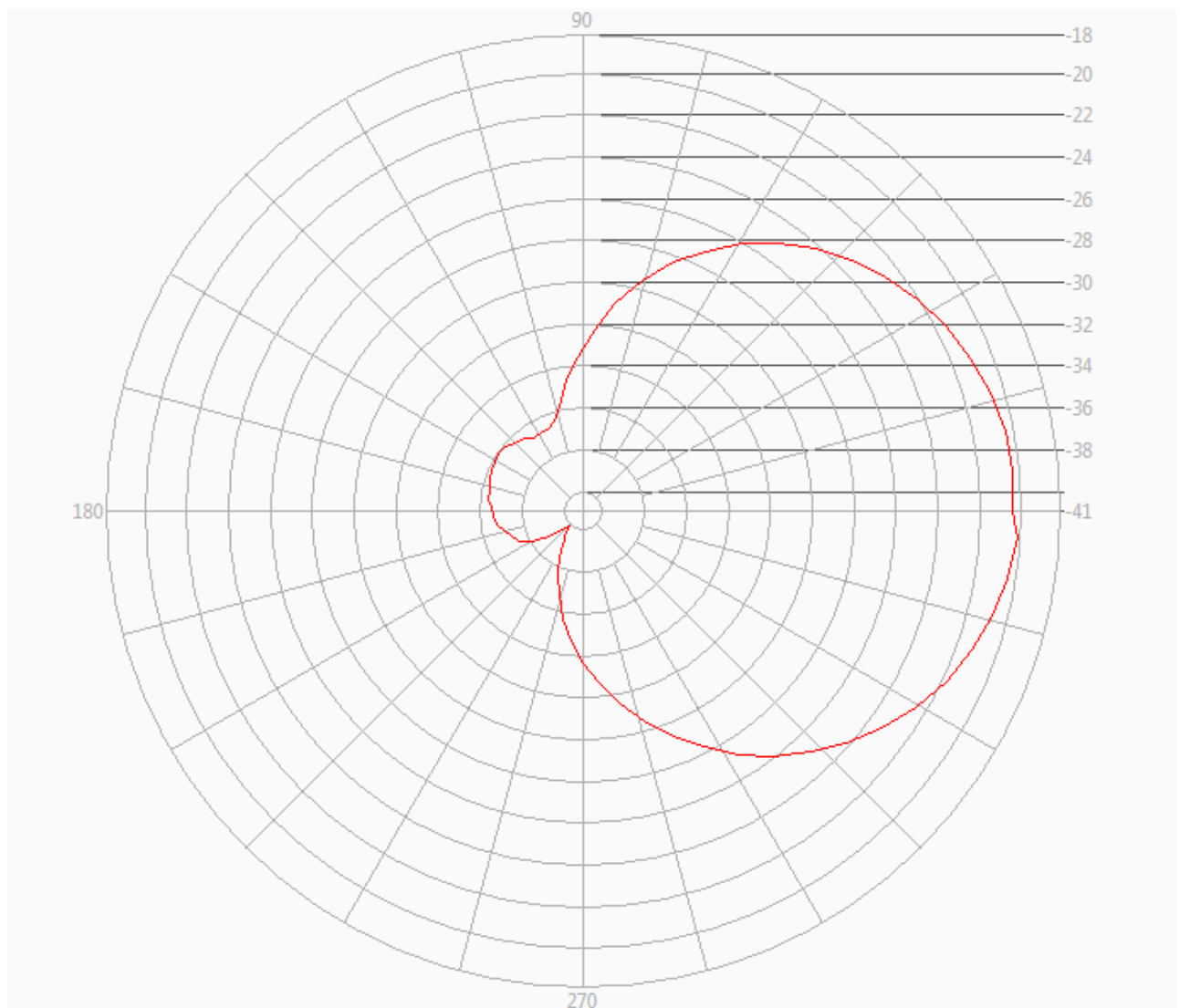
• RADIATION PATTERN



Example 3)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	RMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

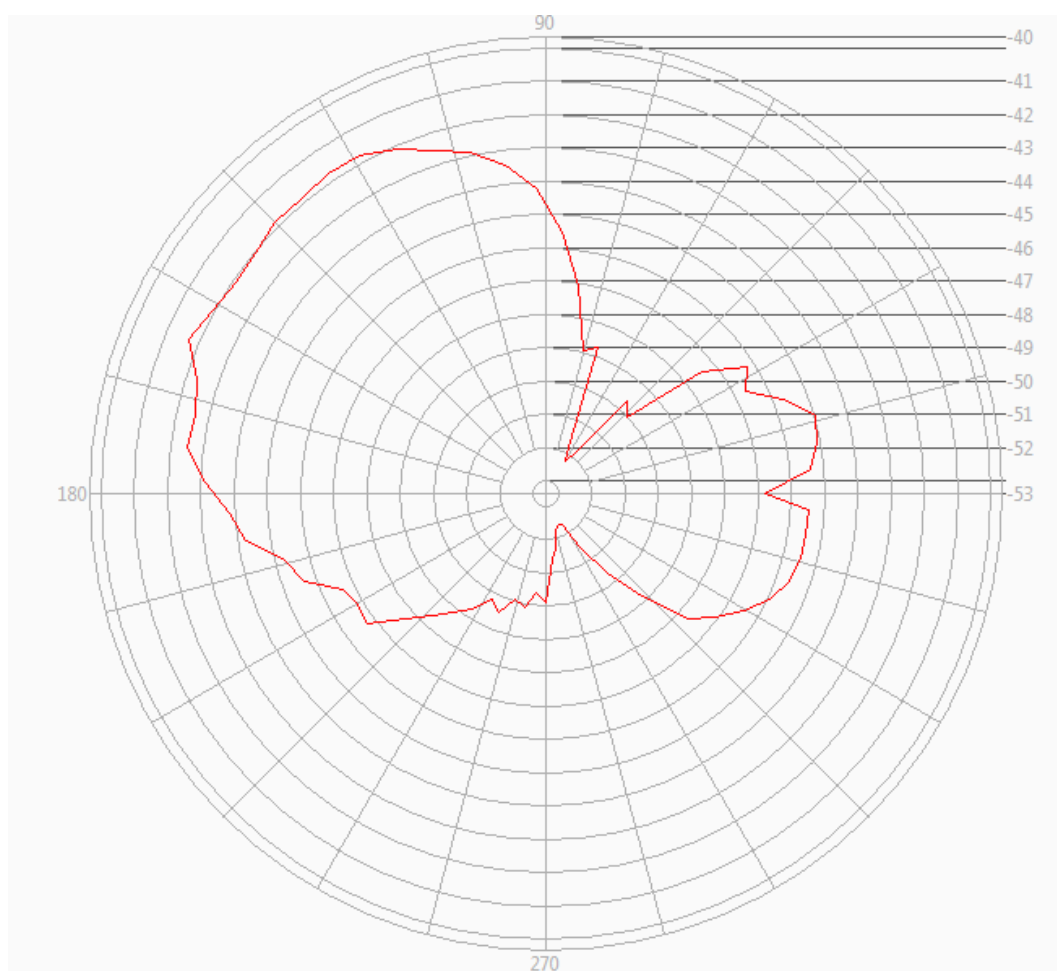
- RADIATION PATTERN**



Example 4)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
CMSA	Angular ring	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

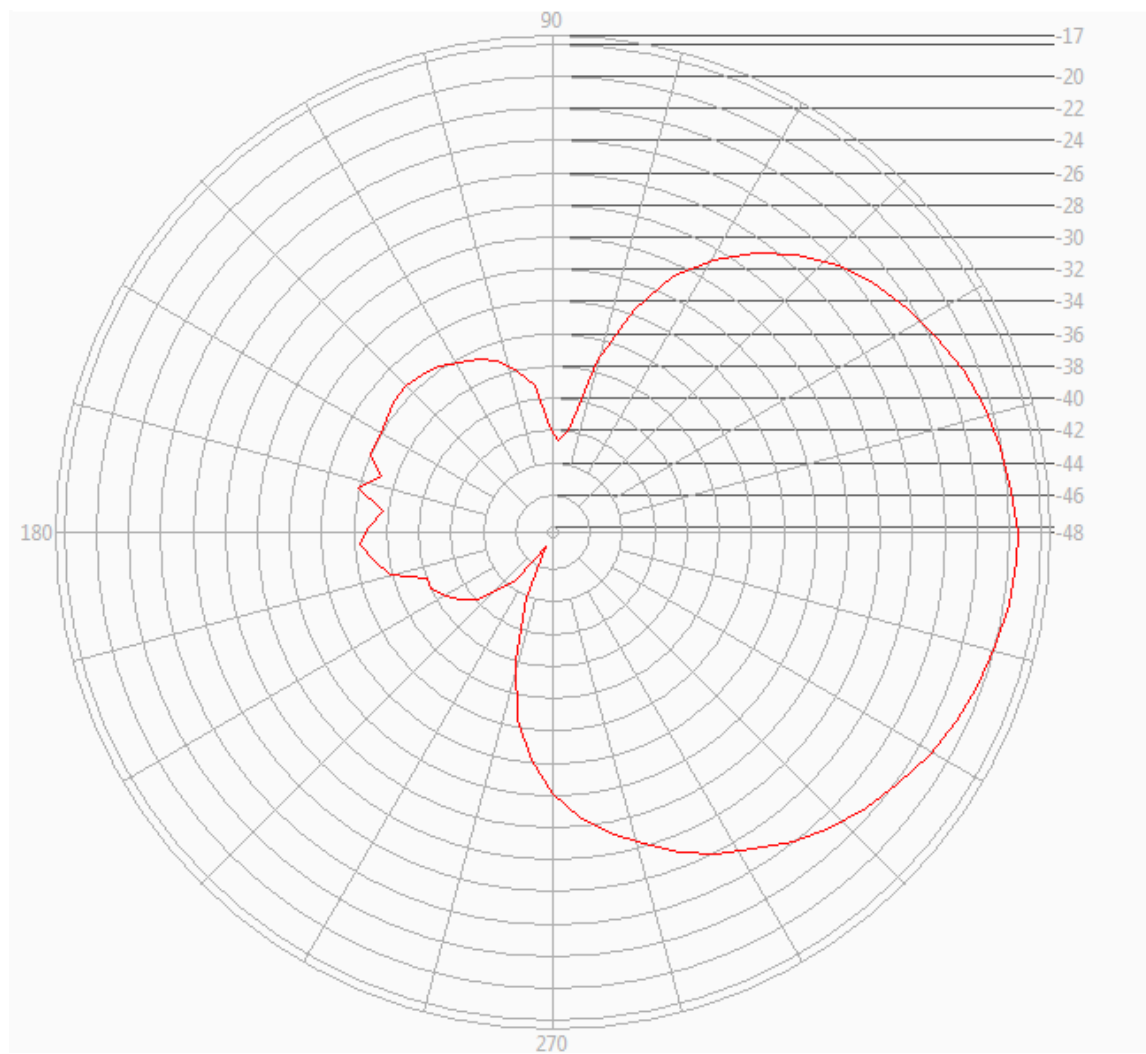
- RADIATION PATTERN**



Example 5)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	CMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

- RADIATION PATTERN**



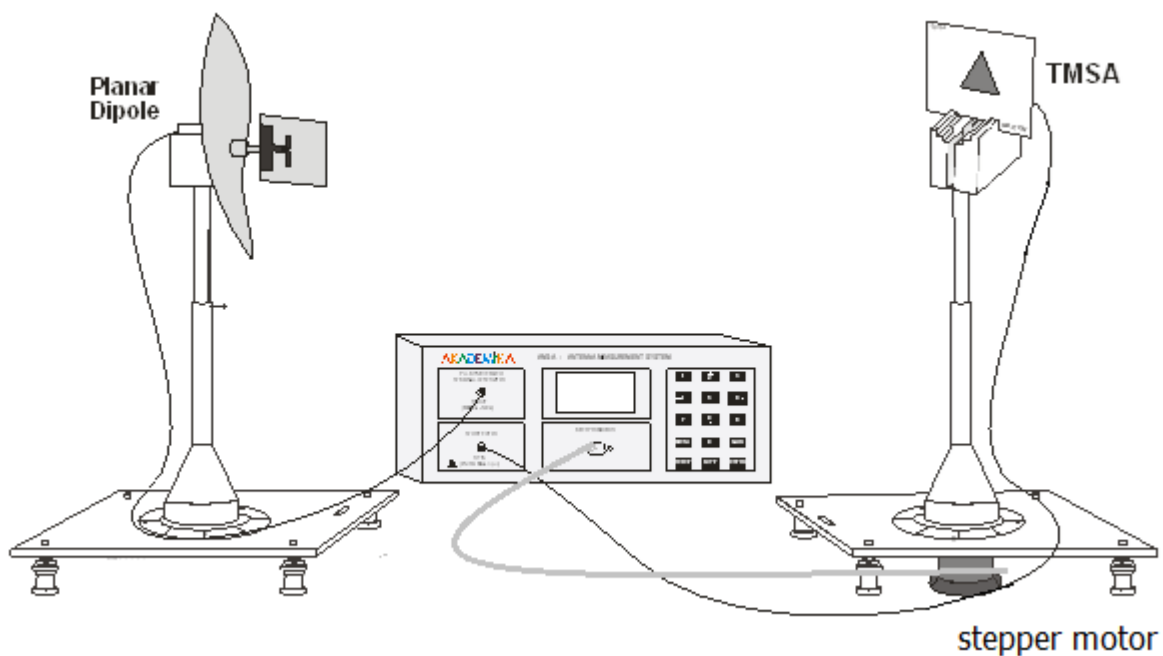
OBJECTIVE

Design and study of basic PLANAR (MICROSTRIP) shapes of antenna and its configurations.

EQUIPMENT REQUIRED:

TX Antenna	RX Antenna -Shapes	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	1. Triangular-TMSA 2. Circular-CMSA 3. Rectangular-RMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:

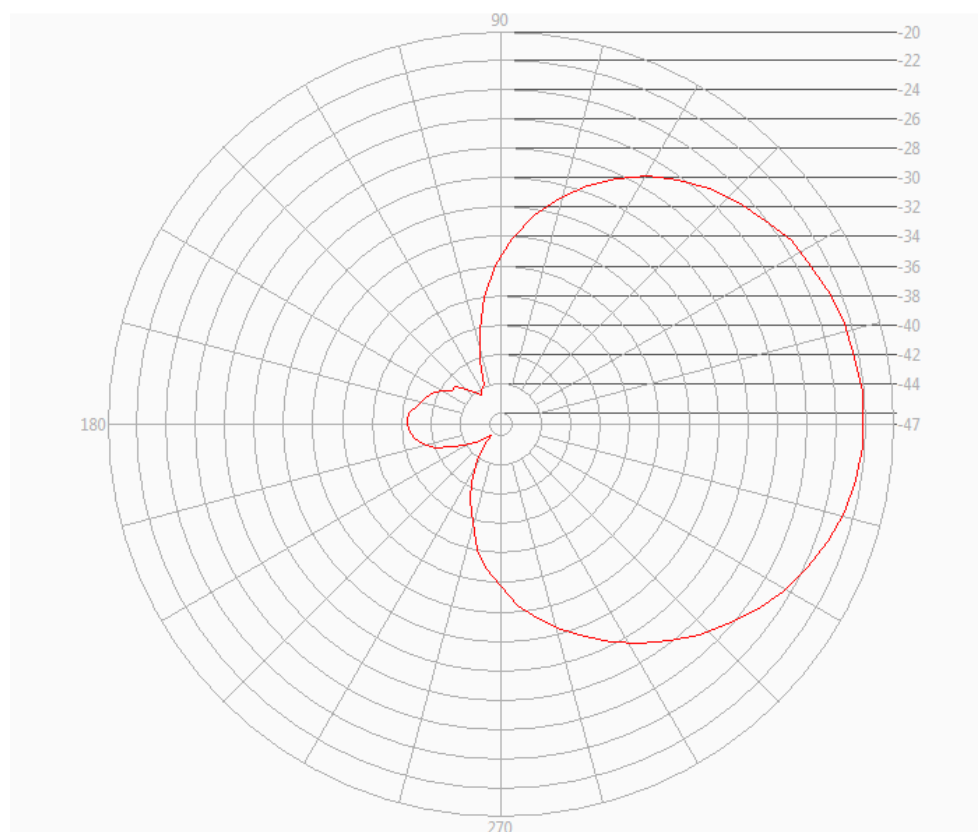


PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **CO-POLARIZATION**, select RX antenna then click on START
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.
(For the formulas used to calculate the above parameters refer Annexure-1)
8. Similarly, repeat the experiment for triangular (TMSA) and rectangular (RMSA) antenna and observe the effect on received power.
9. So from the plots one can find hat which antenna is more directive.
10. From SAVE option, the plot can be saved and take the printout whenever required.

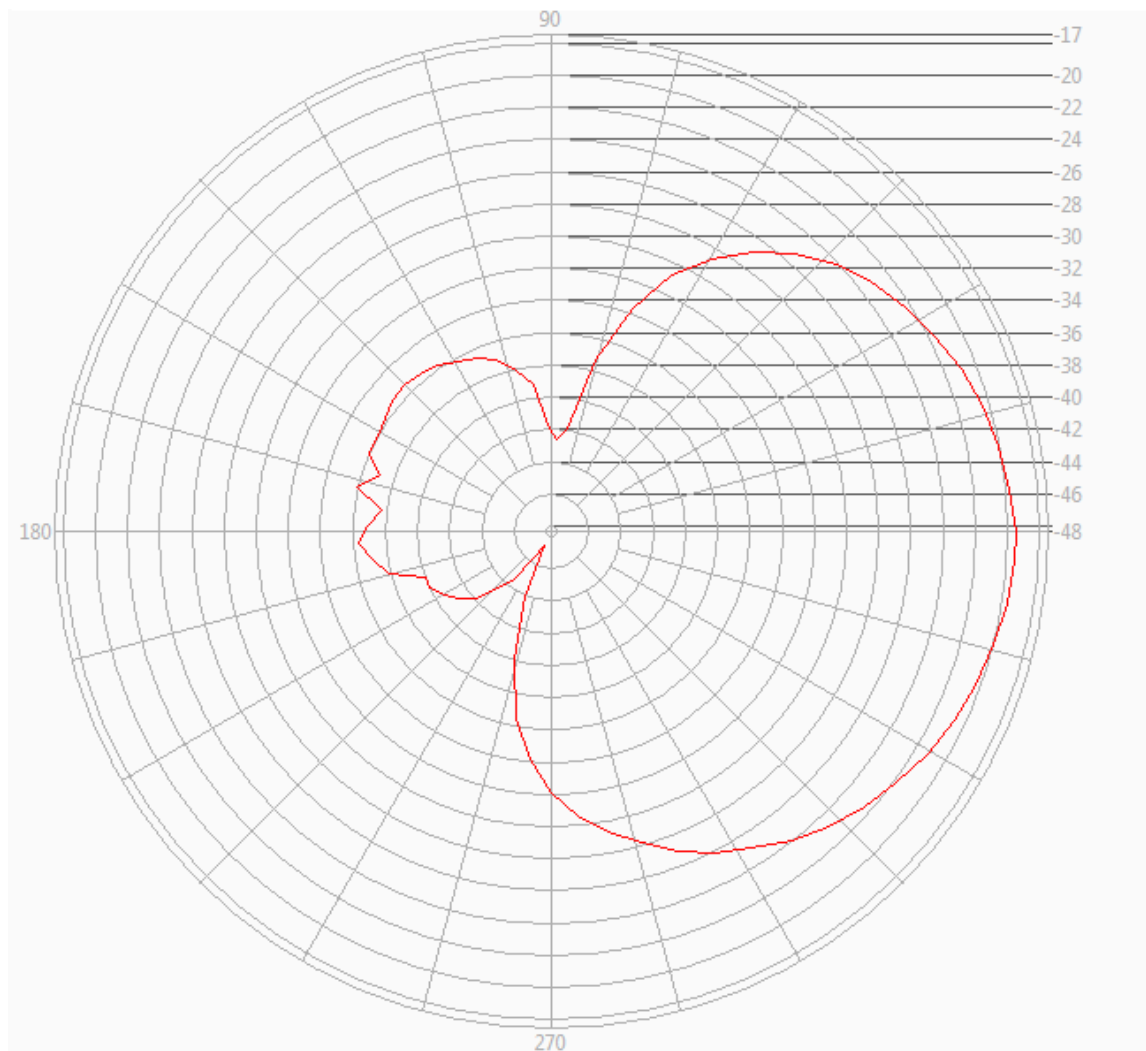
Example 1)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	TMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN

Example 2)

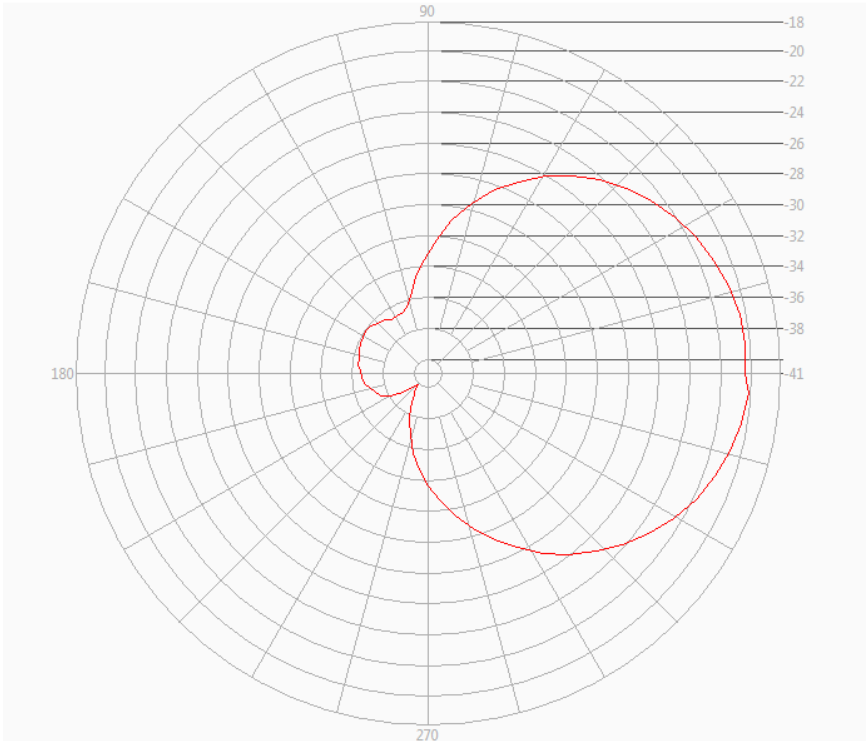
TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	CMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN

Example 3)

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	RMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

RADIATION PATTERN



OBJECTIVE

Effect of different feeding techniques in PLANAR (Microstrip) antenna.

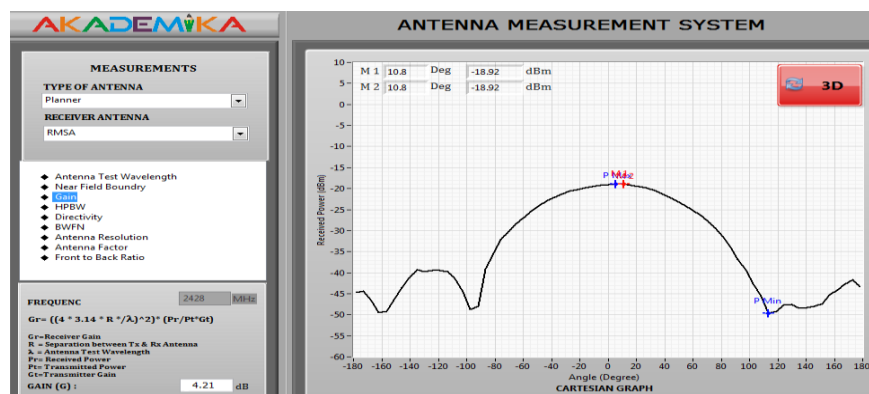
EQUIPMENT REQUIRED:

TX Antenna	RX Antenna -Shapes	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	Any PLANAR (Microstrip) antenna With feedings. 1. Co-axial RMSA 2. Insert feed RMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

PROCEDURE:

1. Set up the experiment as per shown in figure above connect using Co-axial feed.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe files; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **CO-POLARIZATION**, select RX antenna then click on START.
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go in Measurement, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.

(For measurement of above parameter refers Annexure-1)

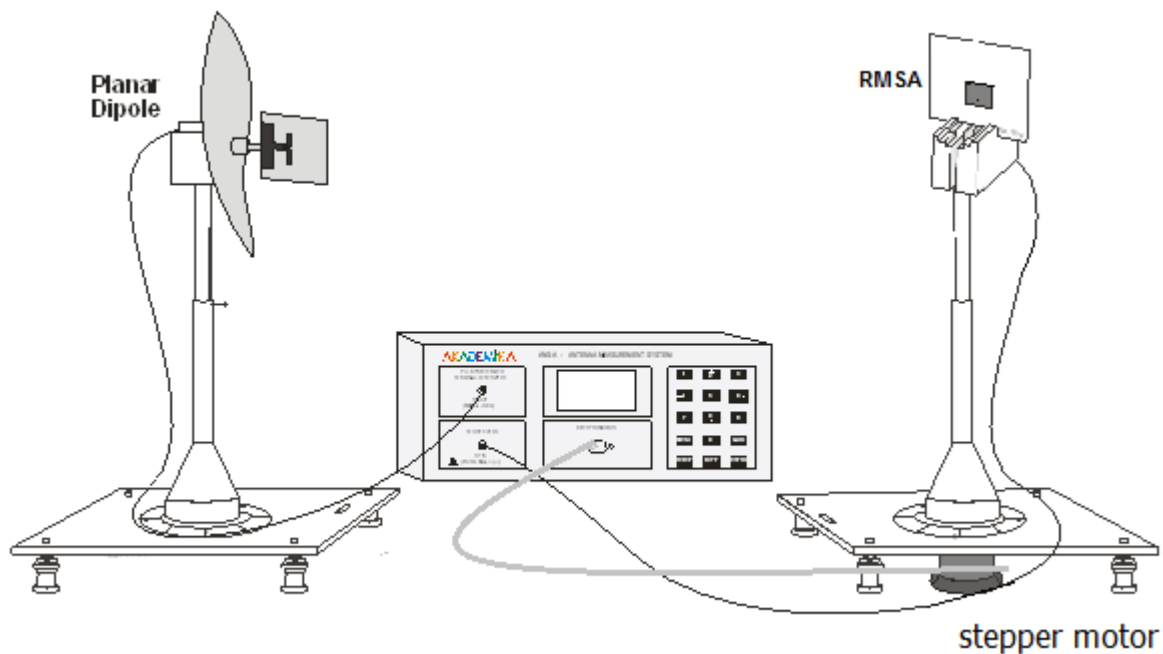


8. Similarly, repeat the experiment for other feeding techniques i.e. insert feed and observe the effect on received power.
9. From SAVE option, the plot can be saved and take the printout whenever required.
- 10.

Example 1) Co-axial feed -RMSA

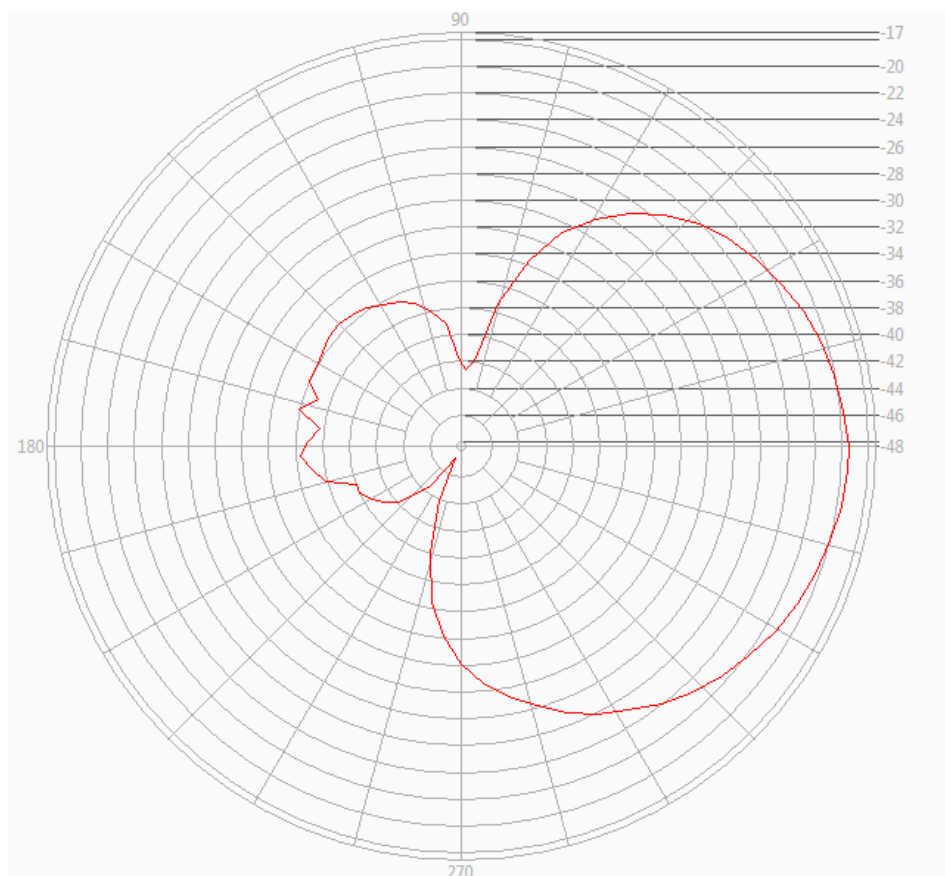
TX antenna	RX Antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	RMSA –Coaxial feed	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



PROCEDURE:

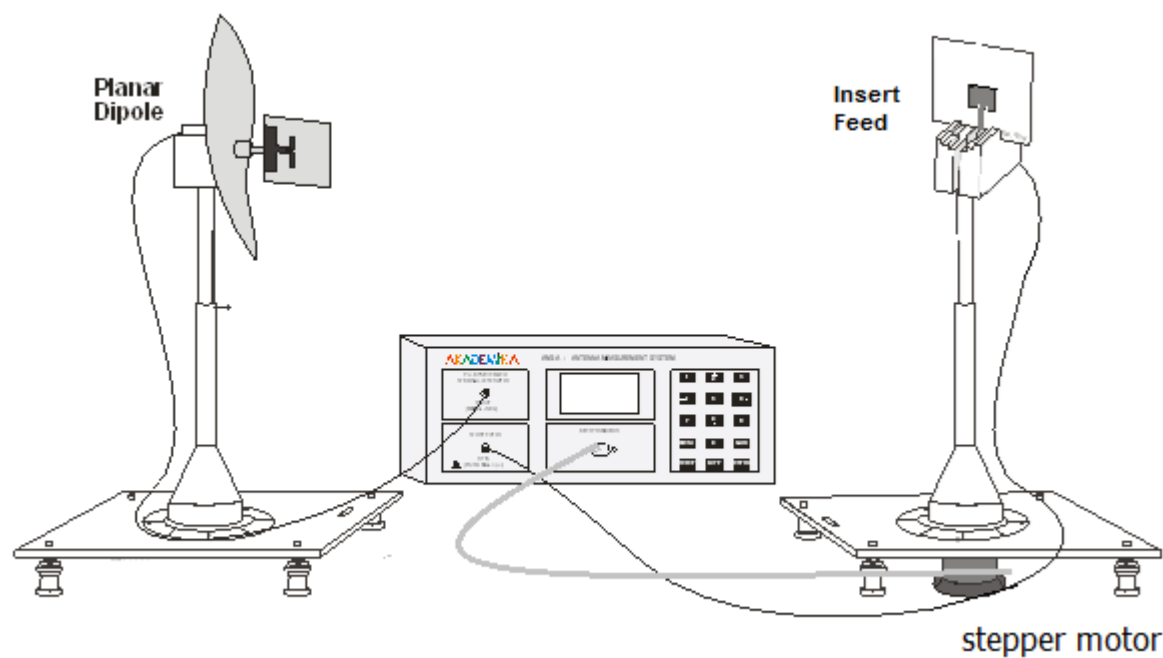
1. Set up the experiment as per shown in figure above connect using Co-axial feed.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe files; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go in **FAR FIELD PATTERN**, select **Front and Back Ratio** – Planer- Insert feed Patch.
6. Then readings from 0 degree to 360 degree will be plotted in the software.
7. Go in Measurement, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.
(For measurement of above parameter refers Annexure-1)

RADIATION PATTERN:

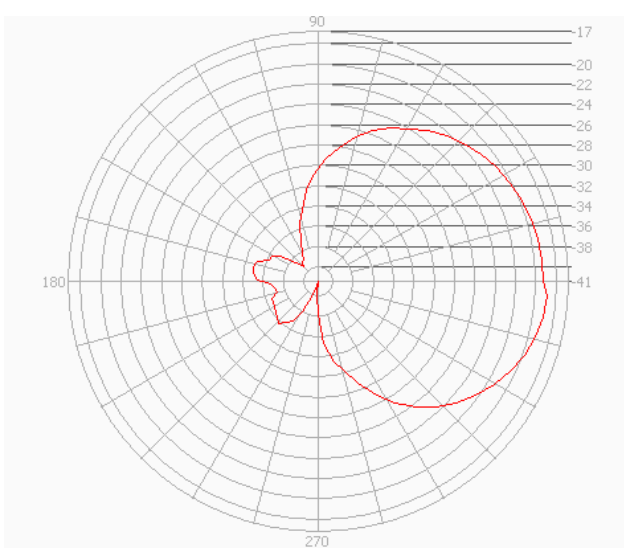
Example 2) Insert Feed Patch

TX antenna	RX Antenna	Transmitter input	Receiver output	CABLE
Planner Dipole	Insert feed Patch	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



RADIATION PATTERN:



EXPERIMENT NO. 10

EXPERIMENT NO. 10

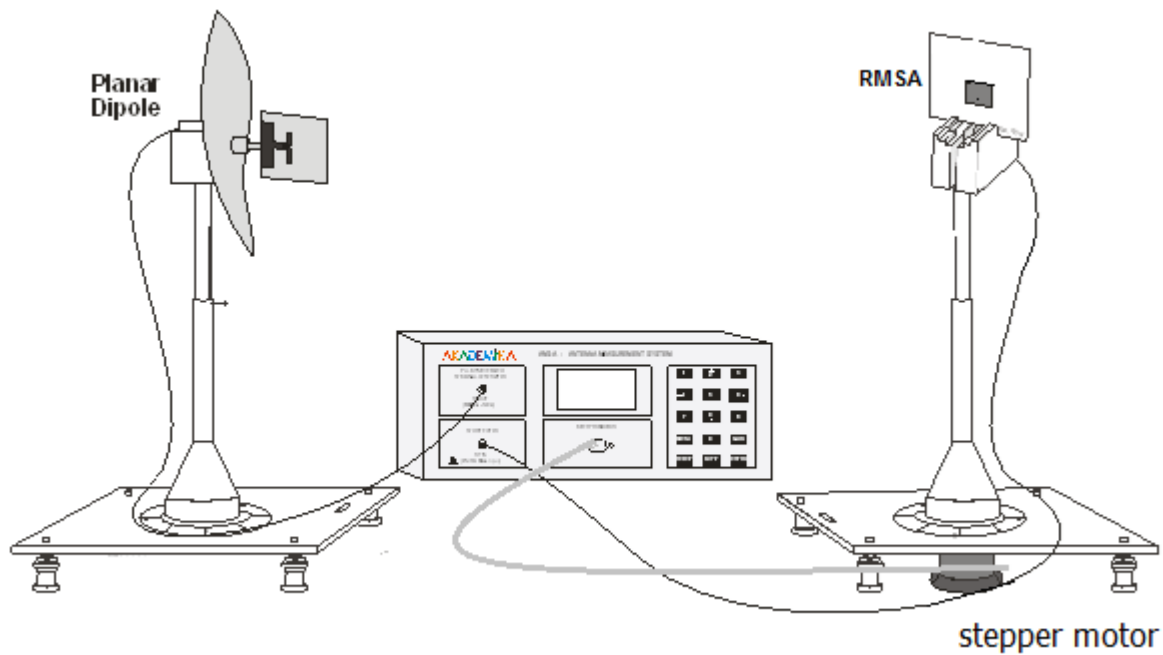
OBJECTIVE

To measure the gain of Microstrip patch Antenna (RMSA).

EQUIPMENT REQUIRED:

TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	RMSA	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

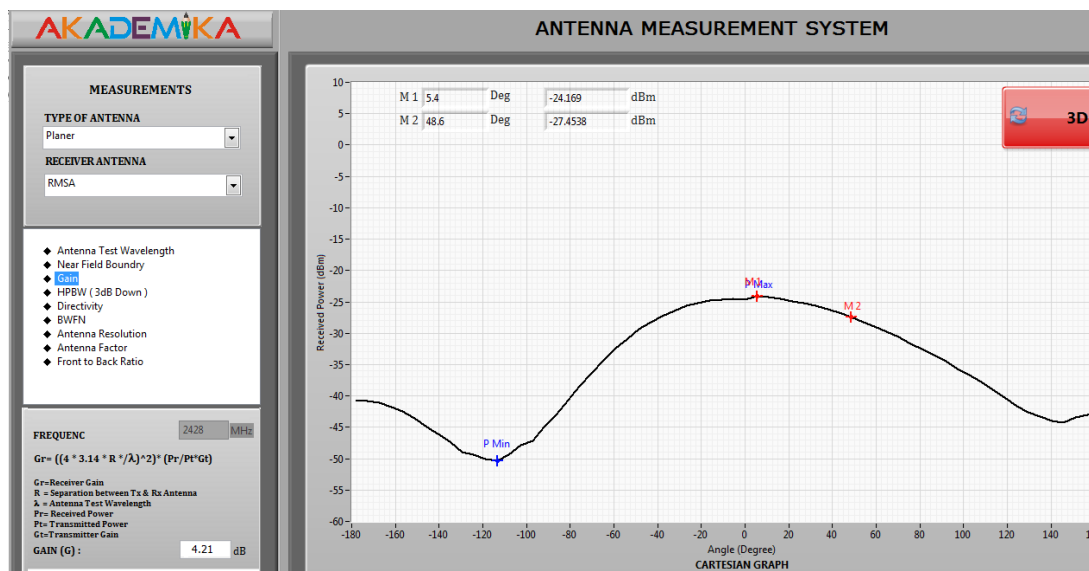
SETUP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file; select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go to Main experiment as **Far field Pattern** and select **“Gain/ Directivity”**
As Experiment, Type of Antenna as **Planner**, Name of RX Antenna as **RMSA**
6. Click on START Experiment.
7. Then readings from 0 degree to 360 degree will be plotted in the software.
8. Go to Measurement tab in the software, select type of antenna, select Receiver antenna then click on Antenna Test wavelength, Near Field Boundary, Gain, HPBW etc.

(For the formulas used to calculate the above parameters refer Annexure-1)



9. From SAVE option, the plot can be saved and take the printout whenever required
10. Refer the same procedure for other antennas.

EXPERIMENT NO. 11

EXPERIMENT NO. 11

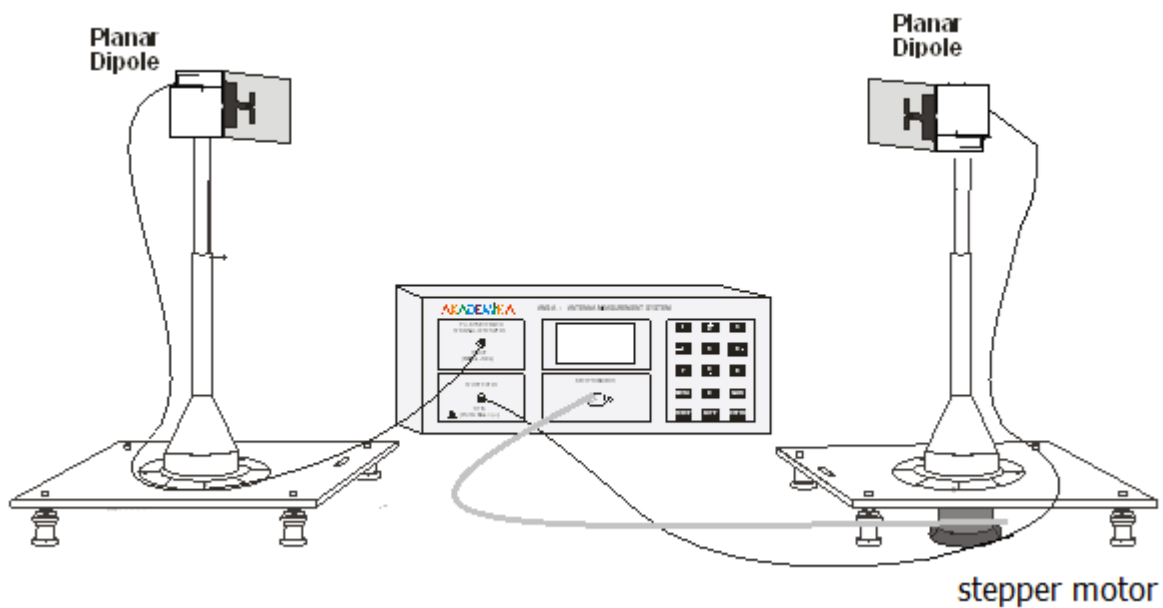
OBJECTIVE

Calibration of Planer Dipole Antenna using Absolute-gain measurements techniques.

EQUIPMENT REQUIRED:

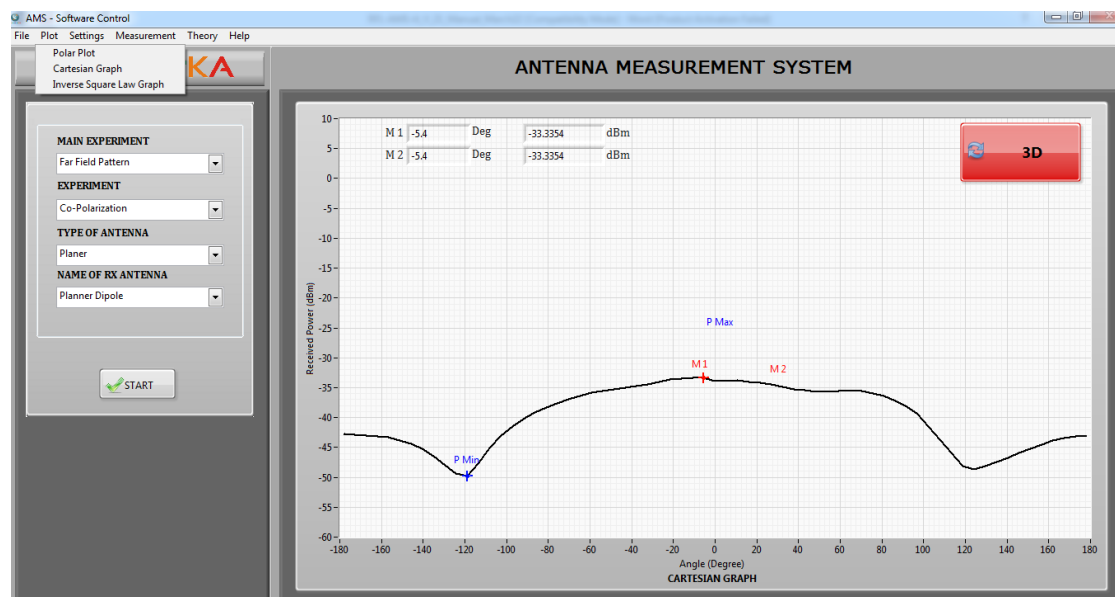
TX Antenna	RX antenna	Transmitter input	Receiver output	CABLE
PLANAR DIPOLE	PLANAR DIPOLE	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

SETUP ARRANGEMENT:



PROCEDURE:

1. Set up the experiment as per shown in figure above.
2. Set the distance between the antennas to be around 60 cm, consult theory for details.
3. Turn ON the module, select control mode.
4. Open the AMS-A.exe file, select the corresponding COM PORT and Click on Run, Now the software will be in running mode.
5. Go to main experiment and select “user defined Expt”
6. Enter frequency as per designed frequency as 2200 MHz
7. Click on start button after fulfillment of instructions (Mount TX & Rx antenna & set Rx antenna 0 degree in co-polarization.)
8. Click yes. Progress bar shows antenna rotation Wait for full rotation.
9. Go to Cartesian plot and note down maximum received Power (Pr).



We got Received Power $P_r = -33\text{dBm}$,

10. Absolute-gain measurements use the Friis transmission formula.

$$(G_A)_{dB} + (G_B)_{dB} = 20 \log \left(\frac{4\pi R}{\lambda} \right) - 10 \log \left(\frac{P_0}{P_r} \right)$$

If the two antennas are identical, it follows that their gains are equal so that

$$(G_A)_{dB} = (G_B)_{dB} = \frac{1}{2} \left[20 \log \left(\frac{4\pi R}{\lambda} \right) - 10 \log \left(\frac{P_0}{P_r} \right) \right]$$

Where $R = 0.6$ m, $\lambda = 0.125$ m, $P_0 = 0$ dBm or 1 mW, $P_r = -33$ dBm or 0.0005 mW

(Reading taken from the Cartesian graph)

Calculate the absolute Gain of planer dipole antenna.

$$G_A = G_B [dB] = \frac{1}{2} [20 \log (4 \cdot 3.14 \cdot 0.6 / 0.125) - 10 \log (1 / 0.0005)]$$

$$= \frac{1}{2} [20 \log (60.288) - 10 \log (2000)]$$

$$= \frac{1}{2} [35.6 - 33.0]$$

$$= 2.6 / 2$$

$$= 1.3 \text{ dB}$$

The Gain of planer dipole antenna is 1.3 dB

EXPERIMENT NO. 12

EXPERIMENT NO. 12

OBJECTIVE

Measurement of user defined antennas and observe its parameters.

EQUIPMENT REQUIRED:

TX antenna	RX antenna	Transmitter input	Receiver output	CABLE
1. WIRE DIPOLE 2. PLANAR (Microstrip) DIPOLE	1. user antenna	Source (RF out)	Detector (RF input)	SMA To SMA cable
QTY =1	QTY =1	QTY =1	QTY =1	QTY =2

Precaution and warning

Do not attach heavy antenna (recommended weight less than 850 gram)

Maximum pattern measurement is possible up to 3GHz.

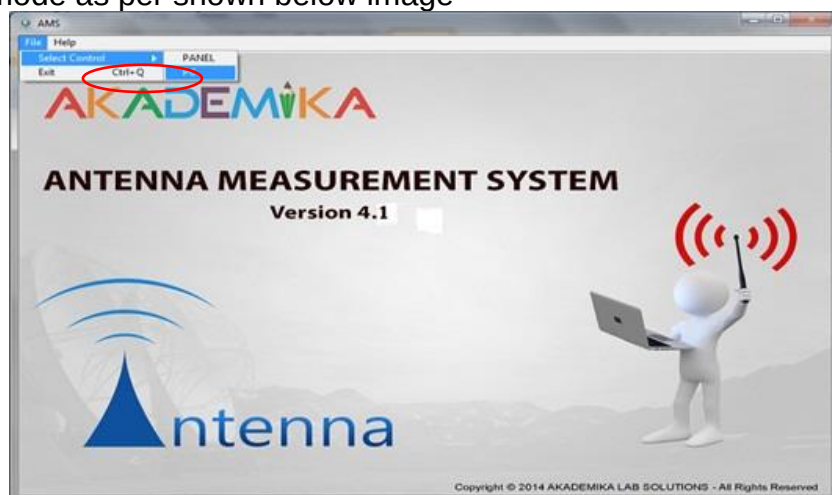
PROCEDURE :

The below procedure is applicable for user designed & fabricated antenna. AMS-A Version 4.1 consist of antenna library, thus software will take care of all necessary settings.

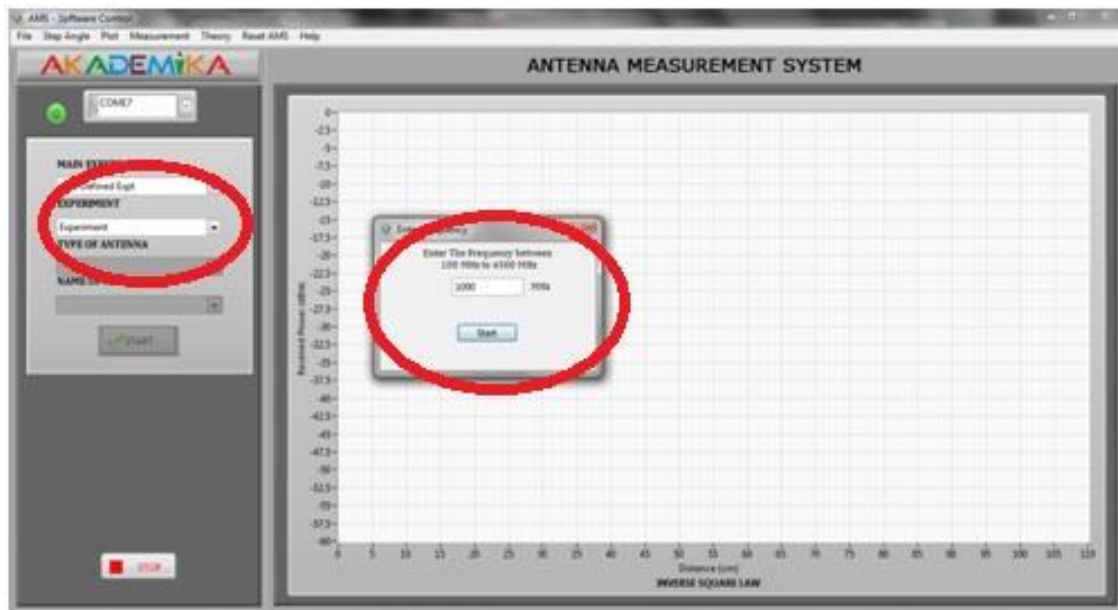
In this procedure we focus on the antenna which is not the part of our library.

Recommend to follow experimental flow for following antennas,

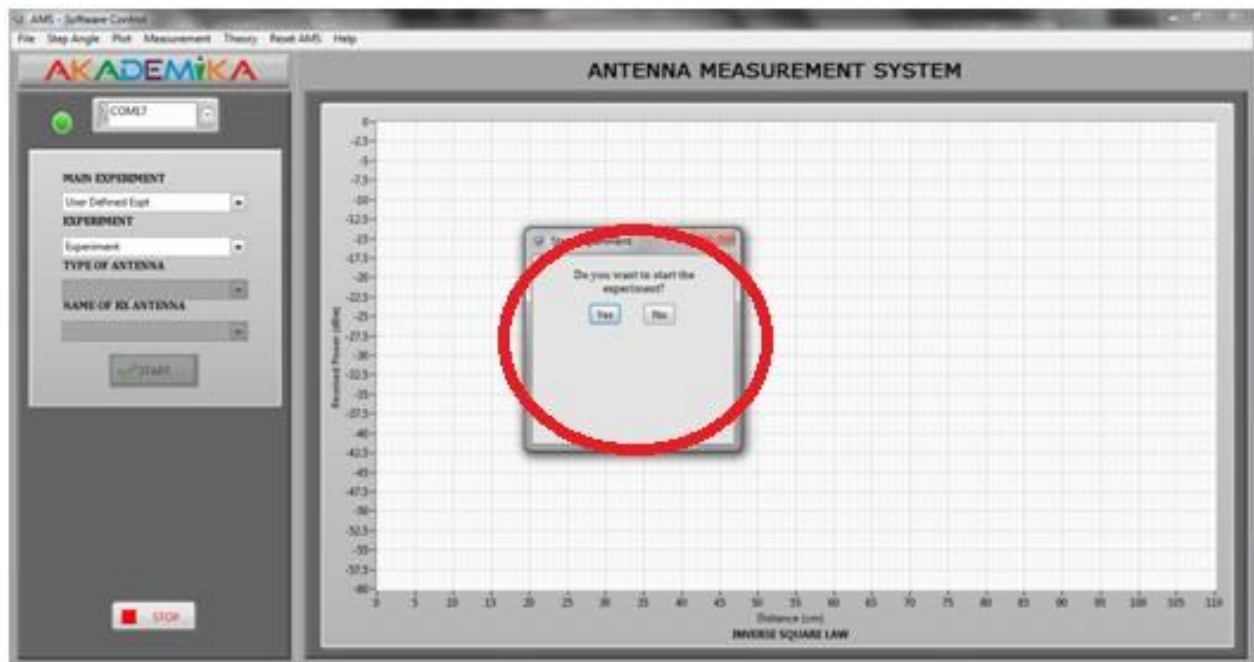
1. Turn on the AMS system→press Menu (on panel)→select pc mode (on panel)→press enter button(on panel)
Panel shows below message
“software mode running”
Click on the AMS logo to start the software.
Select PC mode as per shown below image



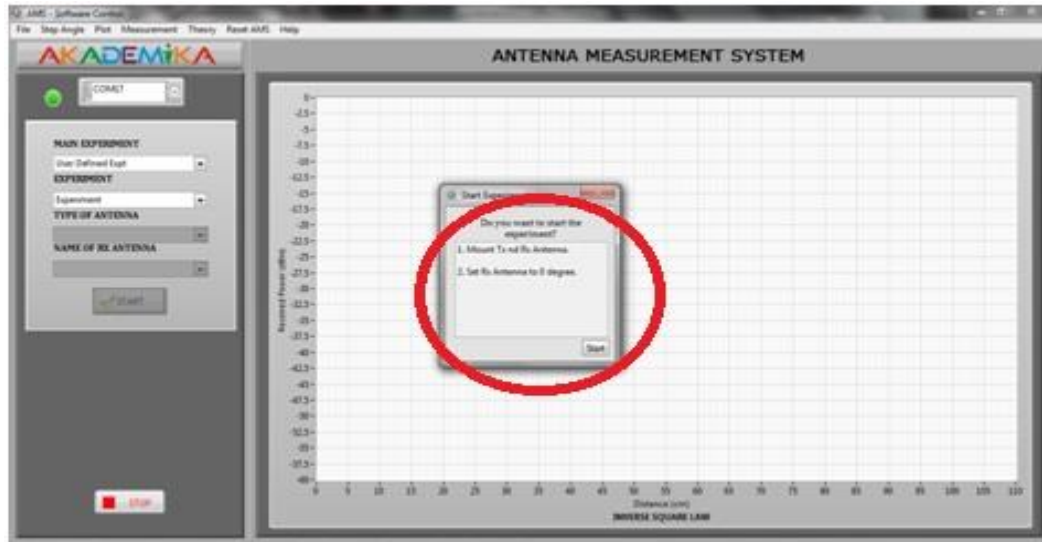
2. Select step angle (1.8deg)
Go to main experiment and select “user defined Expt”
Enter frequency as per your antenna resonance or recommended designed frequency



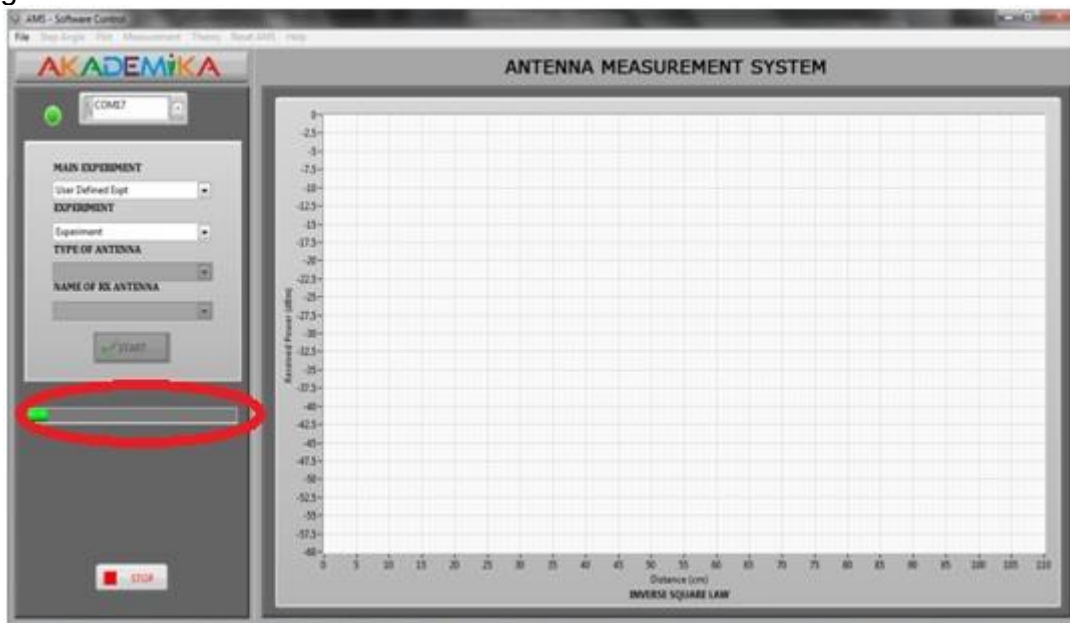
3. Click on start button after fulfillment of instructions (Mount TX & Rx antenna & set Rx antenna 0 degree in co-polarization.)



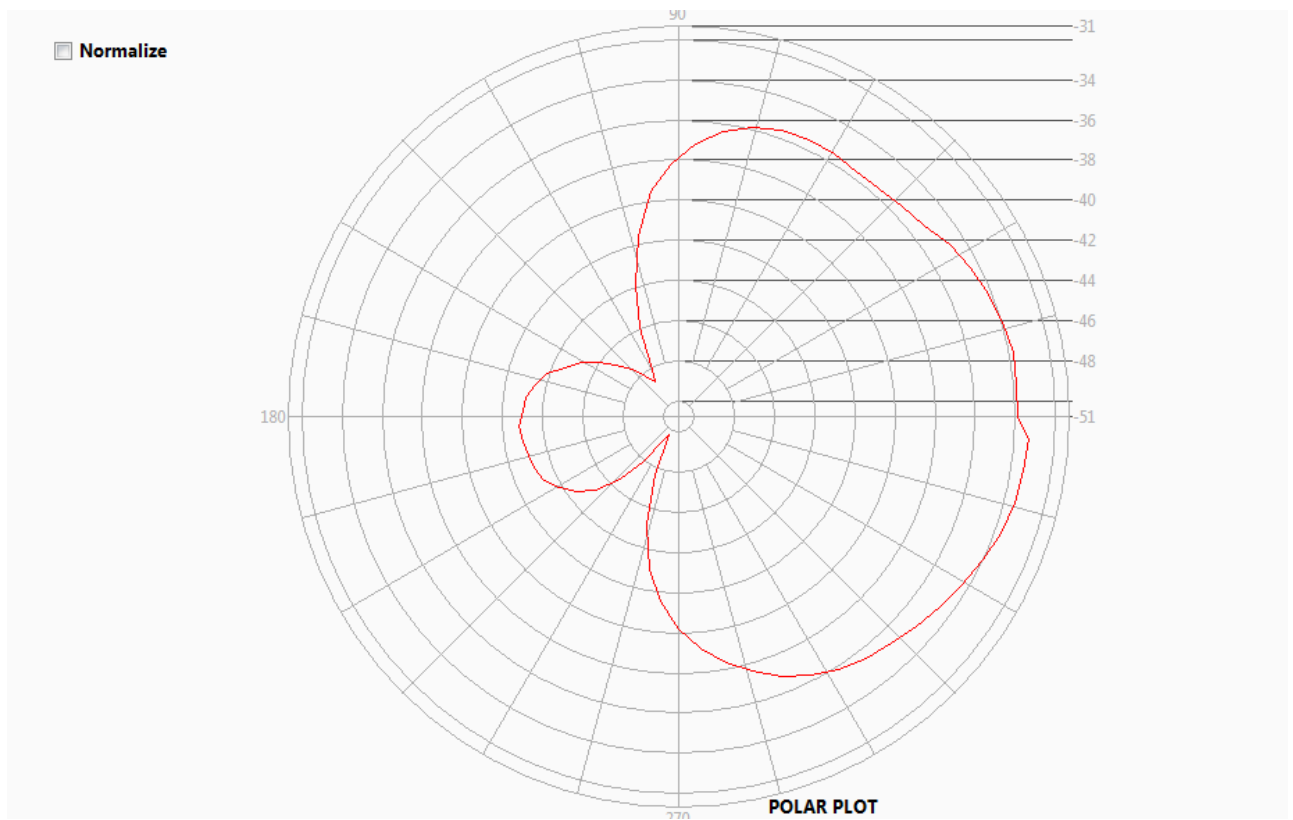
4. Click on **YES** button,



Progress bar shows antenna rotation Wait for full rotation



Pattern will display on polar plot



ANNEXURE 1 : Antenna Calculation Formulae

Antenna Test Wavelength	1) For Wire Antennas, Aperature Antennas, Array Antennas, Reflector Antennas $\text{Antenna Test Wavelength} = (3 * 10^8) / F$ 2) Planer Microstrip Antennas $\text{Antenna Test Wavelength} = (3 * 10^8) / (F * \sqrt{4.4})$
Near Field Boundary	Near Field Boundry = $2 * D^2 / \lambda$ λ = Antenna Test Wavelength D = Diamension of Antenna * Recommended distance between Tx and Rx is 60 cm.
Gain	$G_r = ((4 * 3.14 * R^2 / \lambda^2) * (P_r / P_t * G_t))$ Gr = Receiver Gain R = Separation between Tx & Rx Antenna λ = Antenna Test Wavelength Pr = Received Power Pt = Transmitted Power Gt = Transmitter Gain Lambda = 0.125m, Pt = 0 dBm, R = 0.6m ,Gt= 8 dB, Pr = reading taken from cartesion graph
Half Power Beam Width (HPBW)	M1, M2 = Cursors P1 = Pmax - M1 , P2 = Pmax - M2 HPBW = P1 + P2
Directivity (D)	Directivity = P max / P avg P max = Max Power in mW P avg = Average of All Powers in mW
Beam Width First Null (BWFN)	BWFN = 2*HPBW
Antenna Resolution)	Resolution = BWFN / 1.78 degrees
Antenna factor	$A_f = 9.37 / (\lambda \sqrt{G})$ Where G is Gain and λ is Antenna Test Wavelength
Front to Back Ratio	$\text{Front to Back Ratio} = \frac{\text{Front Field Power}}{\text{Back Field Power}}$

ANNEXURE 2 : Antennas & Their Center Frequencies

Sr No	Transmitting Antenna	Type of Antenna	Receiving Antenna	Center Frequency (MHz)
1	Wire Dipole	Wire Type	Dipole	700
2	Wire Dipole		Helix	1320
3	Wire Dipole		Yagi-Uda 5 element	840
4	Wire Dipole		Rectangular loop	840
5	Wire Dipole		Vee dipole	700
6	Wire Dipole		Folded dipole	700
7	Wire Dipole		Monopole	640
8	Planner Dipole	Aperture Type	Open ended waveguide	2400
9	Planner Dipole		E-Horn	2400
10	Planner Dipole	Planner Type	RMSA	2428
11	Planner Dipole		TMSA	2437
12	Planner Dipole		CMSA	2405
13	Planner Dipole		Annular ring	2445
14	Planner Dipole		2x1 Patch	2392
15	Planner Dipole		Insert Feed Patch	2440
16	Planner Dipole		Chip	2420
17	Planner Dipole		Monopole	1698
18	Planner Dipole		Dipole	2300
19	Wire Dipole	Array	Collinear array	800
20	Wire Dipole		Broadside array	700
21	Wire Dipole	Reflector Type	Corner Reflector	1580

